



A novel hybrid metaheuristic for optimizing deep CNN hyperparameters to enhance heart disease prediction on a comprehensive merged dataset

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Received: 22 June 2025 / Revised: 9 October 2025 / Accepted: 21 December 2025

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Abstract

Early diagnosis of cardiovascular diseases (CVD) is crucial for improving patient survival and reducing healthcare costs. However, the high cost, time-consuming nature of traditional diagnostic methods, and the need for expert interpretation make it difficult to expand early diagnosis, especially in regions with limited healthcare services. In this study, a novel hybrid optimization approach based on Grey Wolf Optimization (GWO), Whale Optimization Algorithm (WOA), and Arithmetic Optimization Algorithm (AOA) is proposed to optimize the hyperparameters of a deep convolutional neural network (CNN) model. The proposed hybrid method operates sequentially in three stages: exploration with GWO, exploitation with WOA, and fine-tuning with AOA. In this study, heart disease prediction is performed using the deep CNN model optimized with the proposed hybrid optimization method on a comprehensive and heterogeneous heart disease dataset (1,190 patient records) generated by merging the Statlog, Long Beach VA, Switzerland, Hungary, and Cleveland datasets. Nine critical hyperparameters, including the number of filters, kernel size, pooling size, fully connected layer neurons, dropout rate, learning rate, mini-batch size, optimizer type, and maximum epoch number, are systematically optimized. The proposed method overcomes the challenges encountered in data integration, preprocessing, and hyperparameter optimization, and provides a prediction model with high accuracy and clinical applicability. Experimental results show that the proposed hybrid optimized deep CNN approach achieves higher accuracy (96.22%), F1 score (96.41%), sensitivity (96.03%), and precision (96.80%) values compared to existing studies. Statistical validation using the Wilcoxon signed-rank test confirmed the significance of the performance improvements ($p=0.016$). SHAP analysis revealed that ST slope, chest pain type, and oldpeak parameters were the most effective features for prediction. Data augmentation performed with SMOTE improved the performance to 97.42% accuracy on the double-balanced dataset. Analyses show that the proposed method provides a practical, explainable and statistically validated tool for early prediction of cardiovascular diseases and helps healthcare professionals make informed decisions.

Keywords Whale optimization algorithm · Heart disease prediction · Hyperparameter optimization · Convolutional neural network · Arithmetic optimization algorithm · Grey wolf optimization

1 Introduction

Early diagnosis of cardiovascular diseases (CVD) is critical for extending patient survival and reducing healthcare costs [1]. Early diagnosis can slow down disease progression, manage treatment processes more effectively, and prevent sudden deaths. However, the high cost of traditional diagnostic methods, time-consuming, and expert interpretation requirements make it difficult to disseminate early diagnosis, especially in regions with limited healthcare resources [2]. This situation reveals the need for fast and

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reliable automated decision support systems. In response to this need, deep learning (DL) and machine learning (ML) based techniques have achieved significant success in the discipline of medical data analysis and prediction of disease in recent years [3]. These methods reduce human error and enable fast and consistent decisions; while offering promising results in early detection of risky individuals, especially in complex and multifactorial diseases such as heart disease [4]. However, most of the existing studies are limited to data sets with limited sample size, usually obtained from a single source; this reduces the generalizability and clinical reliability of the developed models [5].

To tackle issues concerning data volume and heterogeneity, the use of open access heart disease datasets obtained from different sources by merging them has become widespread in the literature [6]. While merged datasets allow training models with more diverse and realistic patient profiles, which also bring new challenges such as differences in feature distributions, missing values, and incompatibility in measurement standards during the data integration process [7]. Therefore, it becomes difficult to produce high-accuracy prediction models from merged datasets without developing effective and systematic data preprocessing and integration strategies.

After these difficulties in data integration and preprocessing steps are overcome, model success largely depends on the correct hyperparameter selection. Determining the hyperparameters (number of layers, learning rate, number of filters, etc.) of deep learning models randomly or by trial and error may cause the model to fall below its potential performance [8]. At this point, the use of meta-heuristic algorithms comes to the fore in order to automate hyperparameter optimization and obtain the best results.

Although metaheuristic algorithms such as Grey Wolf Optimization, Whale Optimization, Arithmetic Optimization Algorithm are promising in hyperparameter optimization, the use of a single algorithm can often lead to problems such as premature convergence, unbalanced exploration/optimization, and getting stuck in local minima [9]. Therefore, in recent years, researchers have turned to hybrid techniques that integrate the capabilities of different techniques. However, studies that systematically assess the performance of hybrid algorithms in hyperparameter optimization of deep learning models, especially on medium-sized and heterogeneous datasets, are still limited [10].

In this paper, a new hybrid method based on Grey Wolf Optimization (GWO), Whale Optimization Algorithm (WOA) and Arithmetic Optimization Algorithm (AOA) is suggested for hyperparameter optimization of deep convolutional neural network (CNN) model on the heart disease dataset combined from open access and different sources. The proposed approach overcomes the existing challenges

in data preprocessing, integration and hyperparameter optimization and provides a high-performance and reliable prediction model in the context of both overall accuracy and clinical applicability.

The contributions of this paper are summarized below:

- **Innovative Hybrid Optimization:** A new hybrid hyperparameter optimization method with sequential three stages (exploration-exploitation-fine-tuning) is developed, combining the strengths of GWO, WOA and AOA algorithms.
- **Dynamic and Optimized CNN Architecture:** The most suitable CNN architecture for the dataset was dynamically created with hyperparameter optimization, and nine critical hyperparameters (number of filters, kernel size, pooling size, number of fully connected layer neurons, dropout rate, learning rate, mini-batch size, optimizer type, and maximum epoch number) were systematically optimized.
- **Comprehensive Comparison:** The proposed approach is extensively compared with existing literature methods in terms of multiple performance metrics (accuracy, F1-score, sensitivity, precision, etc.) and its superiority is demonstrated.
- **Model Explainability:** Using SHAP analysis, the decision-making mechanisms of the model were made explainable, and the effect of clinical parameters (ST slope, chest pain type, oldpeak, etc.) on heart disease prediction was revealed in detail.
- **Data Augmentation Techniques:** With SMOTE-based data augmentation techniques, class imbalance problems were effectively addressed and model performance was significantly improved (97.42% accuracy on the double-balanced dataset).
- **Statistical Validation:** Using the Wilcoxon signed-rank test, it was statistically verified that the performance improvements of the proposed method were not random but represented improvements in true predictive ability ($p=0.016$).
- **Clinical Applicability:** Thanks to its high accuracy and generalizability, it has been shown that the model can be used safely in clinical decision support systems and a practical application example is presented with a user-friendly interface.

2 Literature review

Deep learning and machine learning based approaches in heart disease prediction have recently attracted considerable attention as a research area. In the literature, feature selection, hyperparameter optimization and appropriate

configuration of the algorithm architecture stand out in increasing model success. In particular, meta-heuristic and hybrid optimization techniques make a significant contribution to automatic and effective tuning of model parameters.

In studies using meta-heuristic optimization algorithms, the effect of parameter tuning on model accuracy is remarkable. Dhaka and Nagpal [11] reported 93.26% accuracy in an artificial neural network model optimized with the deer hunting algorithm. El-Shafiey et al. [12] achieved 95.6% accuracy on the Cleveland dataset and 91.4% on the Statlog dataset in the GAPSO-RF approach, where genetic algorithm (GA) and particle swarm optimization (PSO) were integrated with the random forest (RF) model. These results show that determining hyperparameters with systematic optimization algorithms, rather than randomly or by trial and error, increases model success. Saran Kumar and Rekha [13] proposed a stacked ensemble model based on Hawks Optimizer (HO) for heart disease estimation. In the work conducted on the Kaggle cardiovascular disease dataset, an example-based Neural Network Reasoning approach was employed for classification and the overall performance of the approach was optimized with HO.

Similarly, the importance of optimization is emphasized in deep learning-based studies. Mienye and Sun [14] achieved 96.1% accuracy rate on the Cleveland dataset with the stacked sparse autoencoder (SSAE) model optimized with PSO. Modak et al. [15] reported 87.7% accuracy on the combined dataset in the study where they used feature selection and multilayer perceptron (MLP). Wadhawan and Maini [16] achieved 92.81% accuracy on the Cleveland dataset by performing feature selection with improved binary PSO (EBPSO). However, this rate drops to 85.25% on the Long-Beach-Va dataset. This shows that the model success is sensitive to the structure of the dataset and the selected parameters. Rajagopal and Ranganathan [17] evaluated the effectiveness of unsupervised dimensionality reduction techniques in cardiac arrhythmia classification. Five different linear and nonlinear techniques were used in the study: Principal Component Analysis (PCA), Fast Independent Component Analysis (fastICA), Kernel PCA (KPCA), Hierarchical Nonlinear PCA (hNLPCA), and Principal Polynomial Analysis (PPA). The fastICA technique was used with a Probabilistic Neural Network (PNN) classifier on the MIT-BIH arrhythmia database. Saran Kumar and Rekha [18] proposed a technique for cardiovascular disease prediction utilizing Binary Krill Herd metaheuristic optimization (B-KHA) and the CapNet model for feature selection. In the study conducted on the Cleveland dataset, 95% accuracy, 94% specificity, 97% sensitivity, and 95% F1-score were achieved. Saranya et al. [19] proposed a hybrid deep learning approach, DenseNet-ABiLSTM, for multi-class arrhythmia detection employing photoplethysmography

(PPG) signals. The model classifies six different types of arrhythmia using densely connected convolutional networks and Attention-based Bidirectional Long Short-Term Memory (ABiLSTM). An average F1-score of 87.74% and an accuracy of 89.14% were achieved.

It is observed that parameter tuning also increases the model success with classical methods. Mishra et al. [20] achieved 86.89% accuracy in the 1D-CNN model optimized with Grid Search. Rasheed et al. [21] showed the positive effect of hyperparameter optimization on the accuracy rate with the Random Forest model. Warsi et al. [22] achieved 98% accuracy in the SVM model optimized with the genetic algorithm.

Hybrid and ensemble methods are also prominent in the literature. Mienye and Jere [23] achieved high sensitivity and specificity in the XGBoost model with Bayesian optimization. Arunachalam and Rekha [24] developed an ensemble approach for cardiovascular disease prediction that combines the k-Nearest Neighbor base classifier with X-boost, Adaboost, and Random Subspace classifiers. Feature selection was performed using Linear Support Vector Feature Measure (l-SVFM) on the UCI Machine Learning dataset, achieving 95% F-measure, 95% sensitivity, 97% precision, and 96% accuracy. However, this study was limited in terms of generalizability due to the use of a single-source dataset and the lack of testing on heterogeneous data from different geographical regions. Yi et al. [25] reported 96.5% accuracy in the Transformer model optimized with PSO. Alam and Muqem [26] reported accuracy rates of up to 98.99% with a chaotic game optimization-based recurrent neural network. However, these high rates were generally tested on single and homogeneous data sets.

In datasets created by combining different data sources, model success can generally remain at lower levels. The 87.7% accuracy rate obtained by Modak et al. [15] with the combined dataset shows that parameter tuning is even more important in heterogeneous and realistic data structures. Atimbire et al. [27], on the other hand, examined the performance of the approach in the context of various metrics by making comprehensive evaluations on multiple datasets in their study where they performed feature selection with WOA.

In summary, in the literature, in the methods developed for heart disease prediction, feature selection and hyperparameter optimization are decisive in terms of model success; especially meta-heuristic and hybrid algorithms are important in automatic and balanced adjustment of parameters. In many studies, accuracy rates remain in the range of 85–96%, and success rates may be lower, especially in combined and heterogeneous data sets. These findings reveal that in complex and multi-source data sets, systematic and powerful optimization algorithms for determining hyperparameters

are critical for the generalizability and clinical reliability of the model.

The following sections will be presented as follows:

Section 3 describes the detailed methodology of the proposed hybrid GWO-WOA-AOA algorithm and the deep CNN architecture developed for heart disease prediction. In addition, the characteristics of the dataset, preprocessing steps, and the process of performing hyperparameter optimization are detailed.

Section 4 presents the experiments performed and the obtained results; the success level of the suggested approach is evaluated comparatively with existing approaches over different metrics. In addition, the advantages of the proposed hybrid algorithm over classical and modern optimization techniques are discussed.

Section 5 summarizes the general results of the study, emphasizes the importance of the obtained findings in terms

of clinical applications, and offers suggestions for future studies.

3 Proposed methodology

The proposed GWO-WOA-AOA hybrid optimized CNN method presented in Fig. 1 is a comprehensive system developed to provide high accuracy and clinical applicability in heart disease risk prediction. In the first phase of the method, five different open-access heart disease datasets (Cleveland, Statlog, Hungary, Switzerland, and Long Beach VA) were merged to create a heterogeneous dataset consisting of 1,190 patient records. These datasets were standardized using Z-score normalization and divided into three parts: 70% training, 10% validation, and 20% test. The hyperparameters to be optimized were the number of

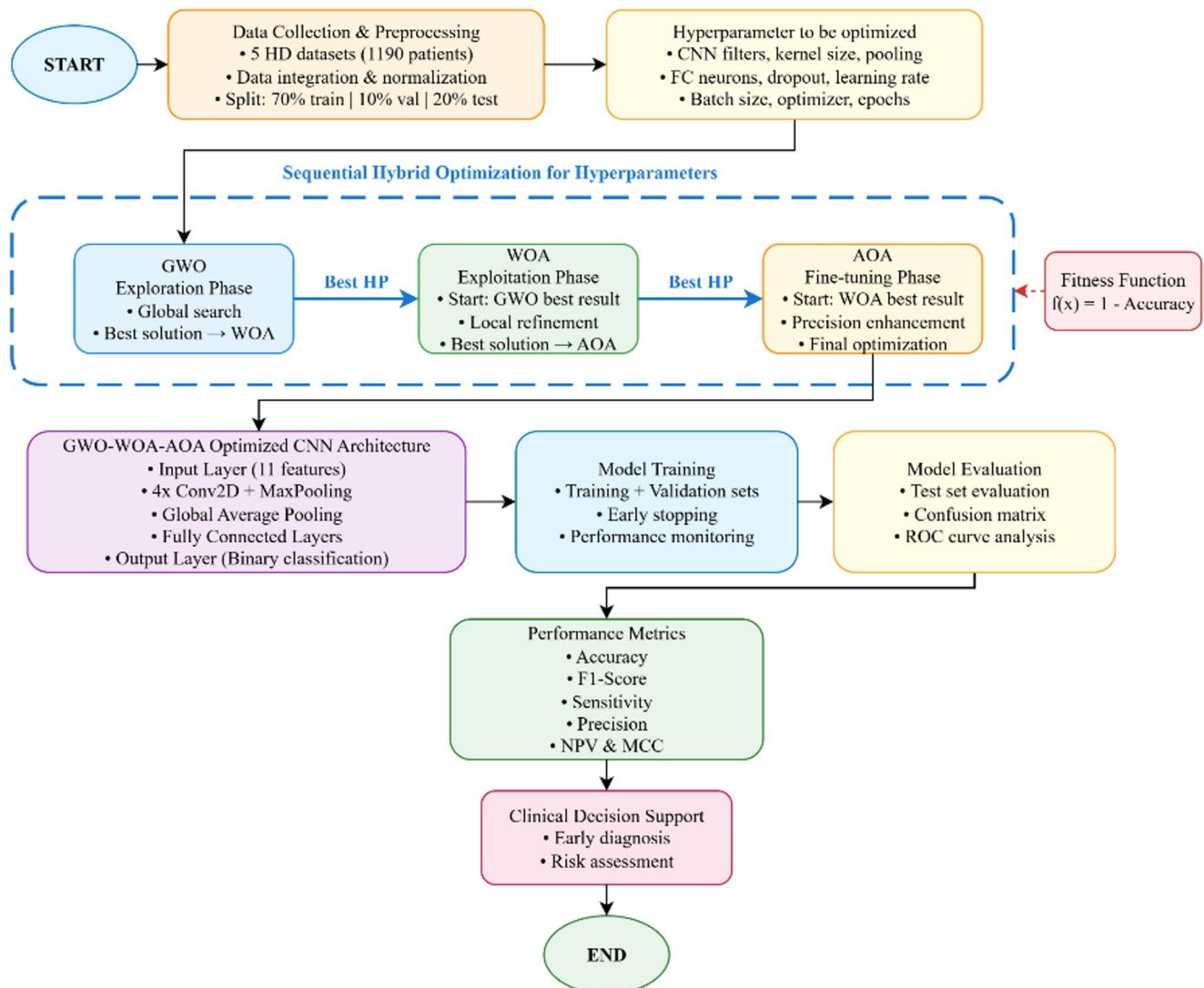


Fig. 1 The flow diagram of the proposed hybrid GWO-WOA-AOA optimized CNN methodology

CNN filters, kernel and pooling sizes, the number of fully connected layer neurons, the dropout rate, the learning rate, the mini-batch size, the optimizer type, and the number of epochs.

The system's unique contribution is the sequential hybrid hyperparameter optimization approach. In this approach, three metaheuristic algorithms operate sequentially: First, the GWO algorithm performs a global search in the large solution space during the exploration phase to identify the best candidate solutions. Second, the WOA algorithm performs local improvements starting from the best solution found by GWO during the exploitation phase. Finally, the AOA algorithm performs precise optimization using the WOA output during the fine-tuning phase to produce the final optimal hyperparameter set. Each algorithm runs 100 iterations, and the fitness function is defined as $f(x) = 1 - \text{Accuracy}$.

The CNN architecture is dynamically generated using the obtained optimal hyperparameters. The model consists of four Conv2D+MaxPooling blocks that process 11 input features, global average pooling layer, dropout layer, fully connected layers, and an output layer for binary classification. Early stopping and performance monitoring techniques are applied during model training.

During the testing phase, model performance is evaluated using accuracy, F1-score, sensitivity, precision, NPV,

and MCC metrics. Model performance is examined in detail using confusion matrix and ROC curve analysis. This method offers a holistic approach encompassing the entire process, from data integration to model evaluation, making a significant contribution to the early diagnosis of heart disease.

3.1 Dataset description

In this paper, the suggested approach for heart disease (HD) prediction was tested on a large HD dataset that is frequently used in the literature and was created by combining different sources. The HD dataset was created by combining five different datasets: Hungarian (294 records), Cleveland (303 records), Statlog (Heart) (270 records), Switzerland (123 records) and Long Beach VA (270 records) [28]. As a result of this integration, one of the largest heart disease datasets in the literature was obtained, consisting of data collected from England, USA, Hungary and Switzerland, containing a total of 1190 patient records. This combination of the dataset provides an important resource, especially for comparing and improving the performance of machine learning approaches. It also contributes to the development of early diagnosis and clinical decision support systems. The dataset includes 11 features including demographic and clinical information of the patients and one target variable (heart disease present/absent). Table 1 summarizes all features in the dataset, their definitions, and value ranges.

3.1.1 Statistical properties of the combined dataset

After merging five different heart disease datasets, comprehensive statistical analysis was performed to understand the characteristics and distribution of the combined dataset. The combined dataset contained 1,190 patient records with 11 clinical features and one target variable, and no missing values were detected in any feature.

Demographic analysis shows that the dataset consisted of 909 male patients (76.4%) and 281 female patients (23.6%), ranging in age from 28 to 77 (mean: 53.72 ± 9.36 years). Age distribution revealed that 41.5% of patients were in the 51–60 age group, 24.8% in the 41–50 age group, 24.6% were over 60, and 9.1% were 40 years of age or younger.

In terms of clinical characteristics, the distribution of chest pain types shows that asymptomatic patients constituted the largest group (52.5%), followed by non-anginal pain (23.8%), atypical angina (18.2%), and typical angina (5.5%). Resting blood pressure values ranged from 80 to 200 mmHg, with a mean of 132.15 ± 18.37 mmHg. Exercise-induced angina was present in 38.7% of patients, while 61.3% had no exercise-induced symptoms.

Table 1 Description of dataset features

Feature No	Feature Name	Definition	Value Range
1	ST slope	ST segment slope during exercise	1: Rising, 2: Flat, 3: Descending
2	Oldpeak	Exercise-induced ST depression (mm)	0.0–6.2
3	Exercise angina	Exercise-induced angina	0: No, 1: Yes
4	fasting blood sugar	Fasting blood sugar (> 120 mg/dl)	0: ≤ 120 mg/dl, 1: >120 mg/dl
5	Resting ecg	Resting ECG results	0: Normal, 1: ST-T abnormality, 2: Left ventricular hypertrophy
6	Max heart rate	Maximum heart rate	60–202
7	Cholesterol	Serum cholesterol (mg/dl)	0–603
8	Chest pain type	Type of chest pain	1: Typical angina, 2: Atypical angina, 3: Non-anginal pain, 4: Asymptomatic
9	Resting bp s	Resting systolic blood pressure (mm Hg)	80–200
10	Sex	Gender	0: Female, 1: Male
11	Age	Patient age (years)	28–77
12	Target	Presence of heart disease	0: No, 1: Yes

Cholesterol levels showed significant variation, ranging from 0 to 603 mg/dl (mean: 210.36 ± 101.42 mg/dl). Fasting blood glucose levels exceeded 120 mg/dl in 21.3% of patients. Maximum heart rate during exercise ranged from 60 to 202 beats/min (mean: 139.73 ± 25.52 beats/min).

ST slope analysis showed that flat slopes (48.9%) and ascending patterns (44.2%) were most common, with descending patterns occurring in 6.8%. Oldpeak values representing exercise-induced ST depression ranged from 0 to 6.2 mm (mean: 0.92 ± 1.09 mm).

The target variable distribution shows a relatively balanced dataset, with 561 patients (47.1%) without heart disease and 629 patients (52.9%) diagnosed with heart disease. This balanced distribution is important because it provides adequate representation for both classes and is advantageous for machine learning model training, reducing the risk of bias towards any one class.

3.1.2 Data preprocessing and normalization

The heterogeneous nature of the dataset, combined from five different sources, necessitated a comprehensive preprocessing strategy. Because these datasets originated from diverse geographic regions and different clinical protocols, special attention was required during the data integration process. To ensure data consistency, the compatibility of feature definitions and measurement units in each dataset was thoroughly examined. During the harmonization process for categorical variables, we verified that variables such as chest pain type, resting ECG results, and ST slope used a consistent coding system across all datasets. The statistical distribution of each feature in the combined dataset was examined, confirming that data from different sources had similar distributional characteristics, demonstrating that there were no systematic differences between the datasets.

As shown in Table 1, the features include age (28–77), exercise-induced angina (0–1), cholesterol (0–603 mg/dl), resting blood pressure (80–200 mmHg), resting ECG (0–2), fasting blood sugar (0–1), maximum heart rate (60–202), oldpeak (0.0–6.2), chest pain type (1: Typical angina, 4: Asymptomatic), gender (0: Female, 1: Male), and ST slope during exercise (1–3).

All categorical variables in the dataset are numerically coded as appropriate. Since the features have different units and ranges, z-score standardization was applied to all input variables so that the model can take each feature into account equally. In this process, the mean of each feature is subtracted and normalized by dividing it by its standard deviation. The normalization formula used is given below:

$$x_{norm} = \frac{x - \mu}{\sigma} \quad (1)$$

where σ is its standard deviation, μ is the mean of the trait of interest, and x is the original value. Thus, a variable with a wide range, such as cholesterol, and a variable with a narrower range, such as age, are brought to the same scale. Finally, the normalized dataset was converted to a four-dimensional input format suitable for the proposed CNN architecture (*feature_num*, 1, 1, *sample_num*).

3.1.3 Correlation analysis

Correlation analysis was performed to better understand the relationships between variables in the data set. The correlation matrix provides important information in terms of both feature selection and model interpretability by revealing the level of linear dependency between variable pairs. Figure 2 presents a heatmap showing the pairwise correlation coefficients between all numerical variables in the data set and the target variable.

When Fig. 2 is examined, it is seen that some variables show a moderate correlation with the target variable. In particular, variables such as maximum heart rate (−0.41), chest pain type (0.46), ST slope (0.51) and exercise-induced angina (0.48) stand out as features that may contribute greatly to predicting the presence of heart disease. In addition, moderate correlations were observed between oldpeak and ST slope (0.52) and chest pain type and exercise angina (0.40). In contrast, it is seen that some variables such as resting ECG and fasting blood sugar have very low correlations with both the target variable and other features.

Correlation analysis is especially important for identifying variables with high correlation and preventing possible multicollinearity problems. This analysis provides guidance both in the feature engineering phase and so as to improve the model's robustness.

3.2 Proposed hybrid GWO-WOA-AOA for hyperparameter optimization

In this section, the role of the suggested hybrid GWO-WOA-AOA technique in optimizing the hyperparameters of the deep CNN model for predicting heart disease is detailed. The basic principles of GWO, WOA and AOA are summarized and it is emphasized that the hybrid approach obtained by combining these algorithms provides a more balanced and effective optimization in the search space. Furthermore, the overall optimization success of the technique is evaluated on CEC 2019 functions; it is shown that it achieves more successful results compared to other methods and the optimization algorithms that constitute it.

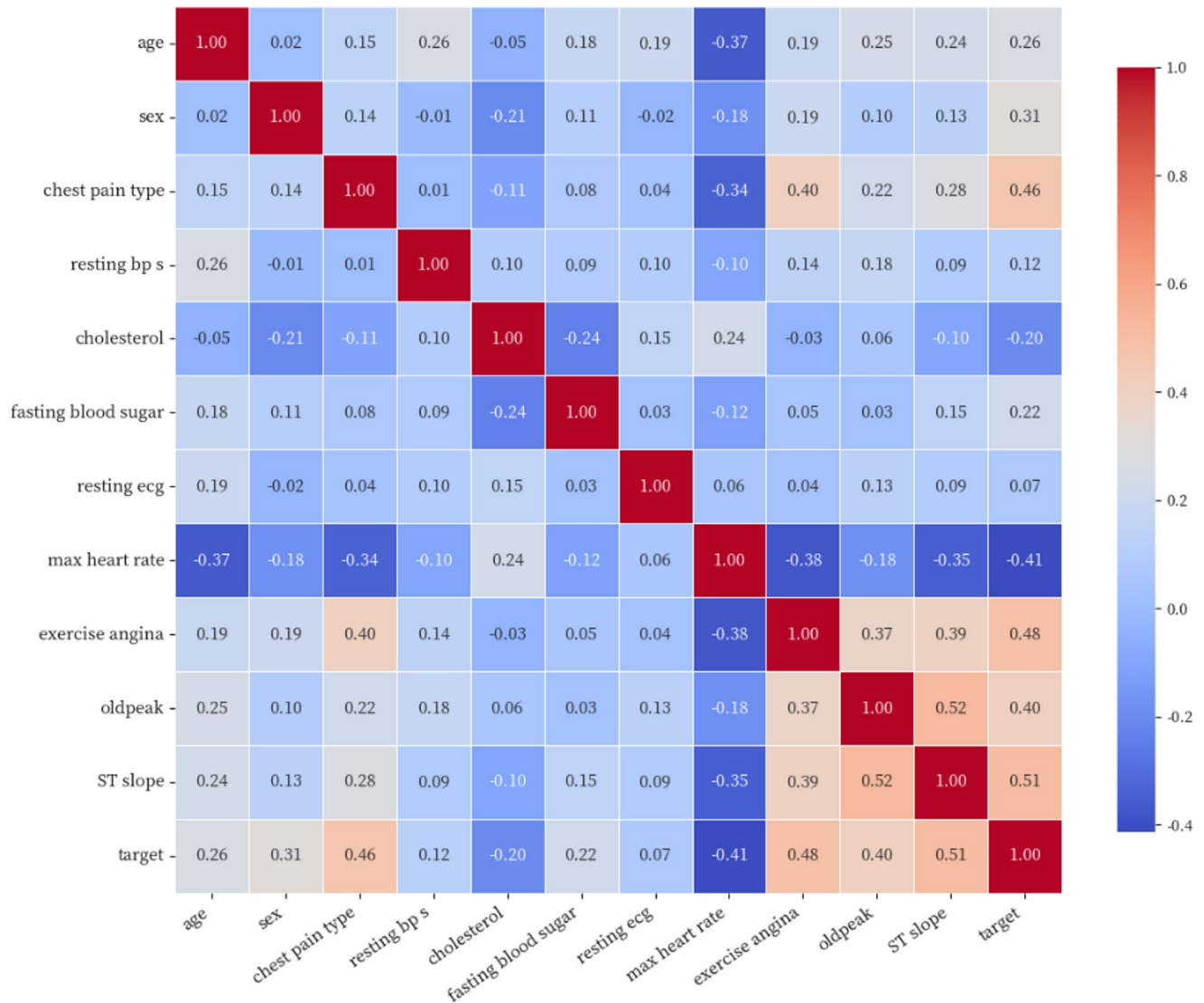


Fig. 2 Heatmap representation of pairwise correlation coefficients between variables in the heart disease dataset and the target variable

3.2.1 Grey Wolf optimization algorithm

The GWO algorithm, proposed by Mirjalili et al. in 2014, is a nature-inspired meta-heuristic optimization approach [29]. Inspired by the group structure and hunting methods of gray wolves, the GWO algorithm represents the social order in the pack at four different levels: alpha (α), beta (β), delta (δ) and omega (ω). In optimization, solutions are hierarchically ranked: the best as alpha, the second best as beta, and the third best as delta, with all others classified as omega wolves [30]. Grey wolves employ a hunting strategy encompassing three primary phases: tracking and approaching the prey, surrounding and immobilizing it, and finally attacking it [29]. These behaviors are formulated in mathematical terms within the GWO. The following equations express prey encirclement behavior:

$$\vec{D} = \left| \vec{C} \cdot \vec{X}_{p(t)} - \vec{X}(t) \right| \quad (2)$$

$$\vec{X}(t+1) = \vec{X}_{p(t)} - \vec{A} \cdot \vec{D} \quad (3)$$

where $\vec{A}$ and $\vec{C}$ are the coefficient vectors, $\vec{X}$ is the current location of the grey wolf, $\vec{X}_p$ is the hunt location (best solution), t is the current iteration. The calculation of these coefficient vectors is performed in the following manner:

$$\vec{A} = 2\vec{a} \cdot \vec{r}_1 - \vec{a} \quad (4)$$

$$\vec{C} = 2 \cdot \vec{r}_2 \quad (5)$$

where $\vec{r}_1$ and $\vec{r}_2$ are random vectors in the interval $[0, 1]$ and $\vec{a}$ is a value that decreases linearly from 2 to 0 over the iteration.

The omega wolves in the GWO algorithm modify their positions by the alpha, beta, and delta wolves, which have more accurate information about where the prey might optimally be located [31]. This process is expressed by the equations presented below:

$$\begin{aligned} \vec{D}_\alpha &= |\vec{C}_1 \cdot \vec{X}_\alpha - \vec{X}|, & \vec{D}_\beta &= |\vec{C}_2 \cdot \vec{X}_\beta - \vec{X}|, \\ \vec{D}_\delta &= |\vec{C}_3 \cdot \vec{X}_\delta - \vec{X}| \end{aligned} \quad (6)$$

$$\begin{aligned} \vec{X}_1 &= (\vec{X}_\alpha - \vec{A}_1 \cdot \vec{D}_\alpha), & \vec{X}_2 &= (\vec{X}_\beta - \vec{A}_2 \cdot \vec{D}_\beta), \\ \vec{X}_3 &= (\vec{X}_\delta - \vec{A}_3 \cdot \vec{D}_\delta) \end{aligned} \quad (7)$$

$$\vec{X}(t+1) = (\vec{X}_1 + \vec{X}_2 + \vec{X}_3)/3 \quad (8)$$

Throughout the iterations of the GWO model, the parameter $\vec{a}$ undergoes a linear reduction from 2 to 0:

$$\vec{a} = 2 - t \cdot (2/T_{max}) \quad (9)$$

Here T_{max} is the maximum number of iterations. The value of the vector $\vec{A}$ controls how the algorithm balances between exploitation and exploration phases [32]: When $|A| < 1$, the wolves advance in the direction of the prey (exploitation); when $|A| > 1$, wolves conduct exploration by moving away from known prey in search of more favorable options. Furthermore, vector C provides random weights of the movement of the prey ($C > 1$ or $C < 1$), contributing to the GWO algorithm's effectiveness at avoiding local optimum entrapment [33]. A detailed flow diagram covering all steps of the GWO algorithm is presented in Fig. 3. This diagram visually details the entire iterative process of the algorithm, including the determination of initial parameters, random initialization of the population, evaluation of the fitness function, parameter updates (a , A , C), and location updates.

In summary, the mathematical model of the GWO algorithm simulates the encircling and hunting strategies of grey wolves using Eqs. (2) and (3), with the coefficient vectors calculated as described in Eqs. (4) and (5). The parameter a is linearly decreased throughout the iterations according to Eq. (9). The position update of omega wolves is guided by the information from the top three solutions (alpha, beta, and delta wolves) as shown in Eqs. (6), (7), and (8). This hierarchical approach ensures a balance between exploitation and exploration, controlled by the magnitude of vector A , while vector C introduces randomness to help protecting the algorithm against entrapment in local optima.

3.2.2 Whale optimization algorithm

A technique that simulates the hunting patterns of humpback whales, the Whale Optimization Algorithm (WOA) stands as one of the nature-based optimization methods [34]. Developed in 2016 by Mirjalili and Lewis, this algorithm mathematically models the hunting strategy of humpback whales by creating a bubble network [35]. WOA is based on three main mechanisms: prey containment, bubble net fishing strategy, and prey search [36]. These mechanisms are represented through the subsequent mathematical formulas: Equations used for prey containment behavior:

$$\vec{D} = |\vec{C} \cdot \vec{X}^* - \vec{X}| \quad (10)$$

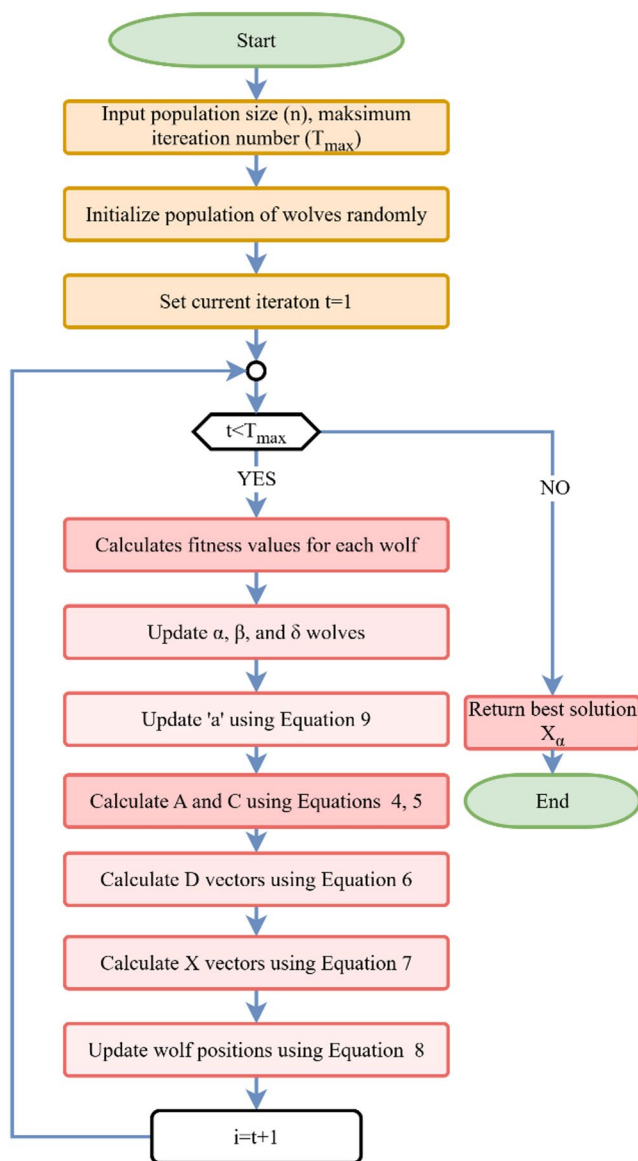


Fig. 3 Flowchart of the GWO algorithm

$$\vec{X}(t+1) = \vec{X}^* - \vec{A} \cdot \vec{B} \quad (11)$$

In Eqs. (10) and (11), $\vec{X}^*$ represents the location of the optimal solution discovered until now, $\vec{X}$ is the current position of the whale and t is the number of iterations [37]. The coefficient vectors $\vec{A}$ and $\vec{C}$, the formula for their computation is presented below:

$$\vec{A} = 2\vec{a} \cdot \vec{r} - \vec{a} \quad (12)$$

$$\vec{C} = 2 \cdot \vec{r} \quad (13)$$

where $\vec{r}$ is a randomly generated vector in the range $[0,1]$ and $\vec{a}$ decreases from 2 to 0 over iterations. The bubble net fishing strategy is modeled by two different approaches [35]:

1. *The shrinking circle approach*: In this approach, the value of $\vec{a}$ decreases, the value of $\vec{A}$ also decreases, and Eq. (10) is used.
2. *Spiral update approach*:

$$\vec{X}(t+1) = \vec{D}' \cdot \cos(2\pi l) + \vec{X}^* \quad (14)$$

$\vec{D}' = |\vec{X}^* - \vec{X}|$ indicates how far the whale is from the prey, b is a fixed value, and l is a random number in the range $[-1,1]$ [36].

Mathematical formulations employed during the algorithm's search phase:

$$\vec{B} = |\vec{C} \cdot \vec{X}_{rand} - \vec{X}| \quad (15)$$

$$\vec{X}(t+1) = \vec{X}_{rand} - \vec{A} \cdot \vec{B} \quad (16)$$

where $\vec{X}_{rand}$ denotes an individual randomly selected from the population [4].

In WOA's operation, a probability value p (usually 0.5) is used to choose between a shrinking circle or a spiral update mechanism. Furthermore, when $|\vec{A}|$ is less than 1, the exploitation process is carried out, and the exploration stage takes place when the magnitude of the vector when $|\vec{A}|$ exceeds 1 [36]. The flow diagram covering all steps of the WOA algorithm is presented in Fig. 4. The diagram visually details the iterative update processes, decision mechanisms, and termination conditions, starting from the initial stage of the algorithm.

3.2.3 Arithmetic optimization algorithm

Inspired by the conduct of mathematical operators, the Arithmetic Optimization Algorithm (AOA) functions as a meta-heuristic approach to optimization [38]. The AOA begins with a randomly created population in its first stage. The procedure for generating the starting population and the ensuing mathematical expression is outlined as follows:

The initial population consists of N solution candidates, and each solution is generated in d -dimensional space using the equation detailed below:

$$X_i = LB + rand(0,1) \cdot (UB - LB), i = 1, 2, \dots, N \quad (17)$$

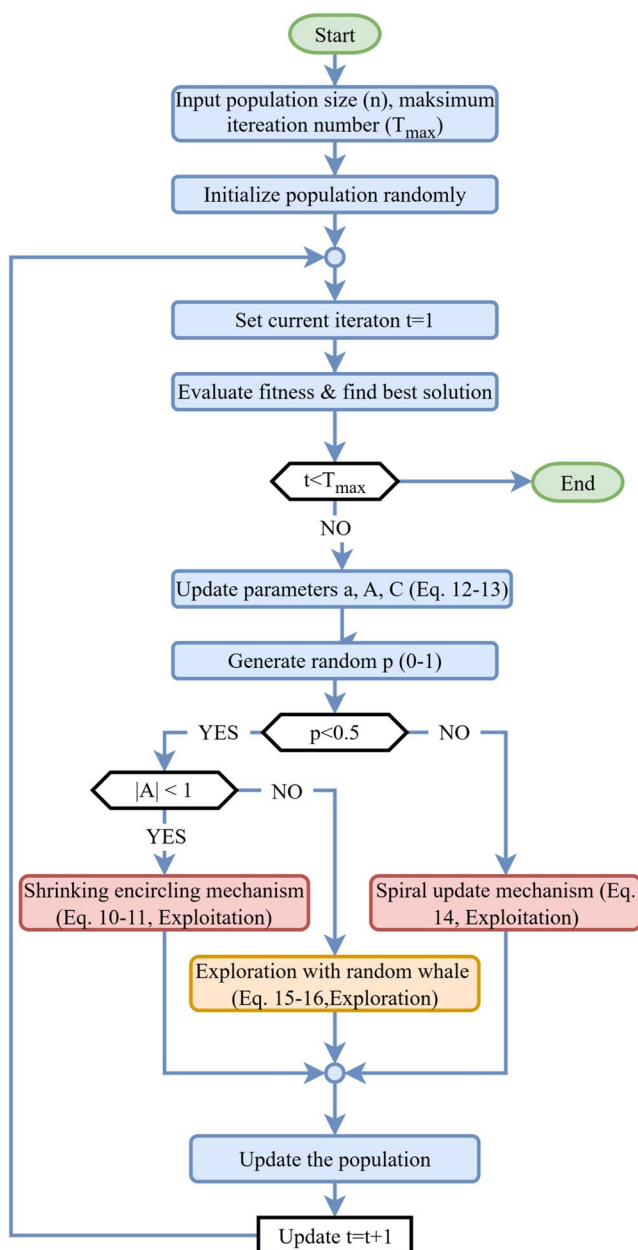


Fig. 4 Flowchart of WOA algorithm

In Eq. (17), $\text{rand}(0,1)$ is the random number generated in the interval $[0,1]$, UB and LB are the upper and lower bounds, respectively and X_i is the i . solution candidate. Once the initial population is created, the Mathematical Operator Proportion (MOP) is calculated at each iteration of the algorithm:

$$MOP(t) = 1 - \left(\frac{t}{T_{max}} \right)^\alpha \quad (18)$$

In Eq. (18), T_{max} indicates the maximum number of iterations, t denotes the current iteration, and α the control parameter. The MOP value regulates the transition amid the algorithm's phases of exploitation and exploration. The algorithm is divided into two phases by comparing the random number r_1 with MOP. In the case of $r_1 > MOP$, the

local search phase is active, and the equations presented below are employed:

$$X_i^{t+1} = \frac{X_{best}}{MOP} \cdot ((UB - LB) \cdot \mu + LB), \text{ for } r_3 < 0.5 \quad (19)$$

$$X_i^{t+1} = X_{best} \cdot MOP \cdot ((UB - LB) \cdot \mu + LB), \text{ for } r_3 \geq 0.5 \quad (20)$$

If $r_1 \leq MOP$, the fine-tuning phase is activated:

$$X_i^{t+1} = X_{best} - MOP \cdot ((UB - LB) \cdot \mu + LB), \text{ for } r_2 < 0.5 \quad (21)$$

$$X_i^{t+1} = X_{best} + MOP \cdot ((UB - LB) \cdot \mu + LB), \text{ for } r_2 \geq 0.5 \quad (22)$$

The parameters r_1 , r_2 , and r_3 , are randomly generated values in the interval $[0,1]$. Equations (19) and (20) provide an efficient exploration of the solution space throughout the local search process, while Eqs. (21) and (22) improve the ability to converge to the local optimum in the fine-tuning phase. All the steps of the AOA algorithm are shown in Fig. 5. As observed in Fig. 5, the algorithm begins with random population generation, followed by the calculation of the MOP at each iteration. Based on the randomly generated values, the Fine-Tuning Phase is activated, and the solution positions are updated. This process continues until the maximum number of iterations is reached, and the best solution is obtained.

In summary, the AOA algorithm initiates by generating a random population as described in Eq. (17), and then iteratively updates the solutions based on the MOP calculated by Eq. (18). The algorithm alternates between exploration and exploitation phases by comparing a random value with MOP. When the local search phase is triggered ($r_1 > MOP$), the positions of the solutions are updated employing Eqs. (19) and (20) to enhance exploration capabilities. Conversely, if the fine-tuning phase is activated ($r_1 \leq MOP$), Eqs. (21) and (22) are employed to improve convergence towards the optimum. The random parameters r_1 , r_2 , and r_3 ensure diversity in the search process.

3.2.4 Proposed of hybrid GWO-WOA-AOA technique

The flow diagram summarizing all stages and general working principle of the hybrid GWO-WOA-AOA technique, which aims to provide a more effective and balanced optimization process by combining the advantages of three different meta-heuristic methods, is presented in Fig. 6. This hybrid structure consists of three main stages: exploration, exploitation and fine-tuning.

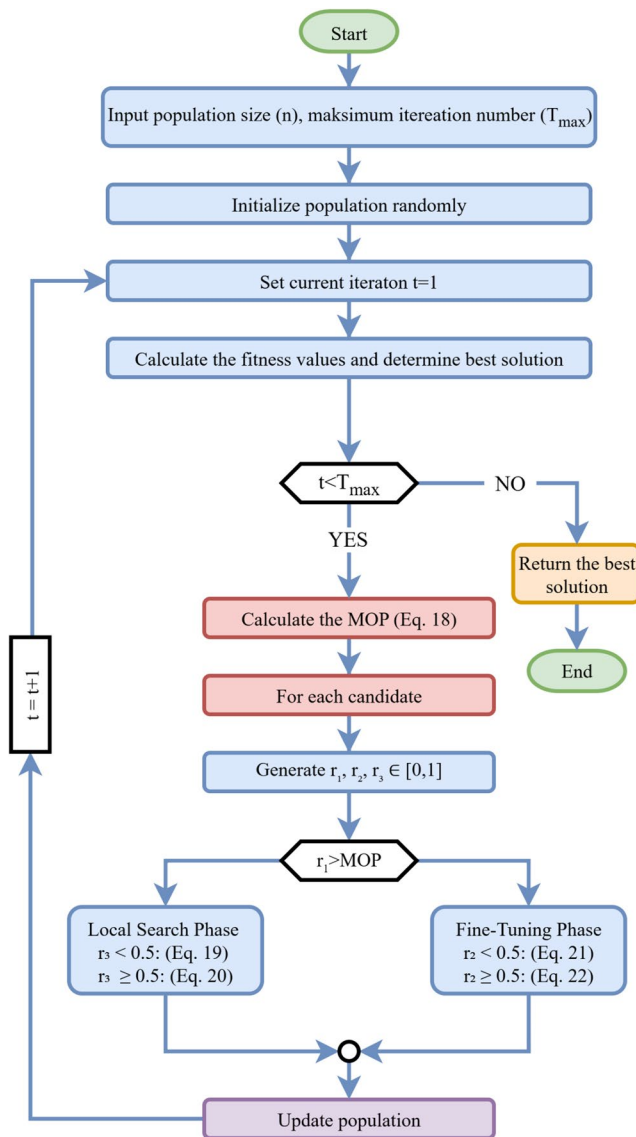


Fig. 5 Flowchart of AOA algorithm

Pseudo-code for algorithm GWO-WOA-AOA	
<pre> INPUT: population_size_gwo; population_size_woa; population_size_aoa; max_iterations_gwo; max_iterations_woa; max_iterations_aoa; dimension; bounds[lower, upper]; objective_func; OUTPUT: best_solution_aoa BEGIN GWO-WOA-AOA // Phase 1: Grey Wolf Optimizer (Exploration) Initialize_Wolf_Population(population_gwo, 20); best_solution_gwo = Find_Best(population_gwo) FOR iteration = 1 to max_iterations_gwo DO a = 2 - iteration * (2 / max_iterations_gwo) FOR each wolf in population_gwo: Update_Leadership_Hierarchy() Calculate_A_and_C_Vectors() Update_Wolf_Position(wolf, α, β, δ, a) END FOR Update_Best_Solution_GWO() END FOR // Phase 2: Whale Optimization Algorithm (Exploitation) Initialize_Whale_Population(population_woa, 20, best_solution_gwo) best_solution_woa = best_solution_gwo FOR iteration = 1 to max_iterations_woa DO a = 2 * (1 - iteration / max_iterations_woa) FOR each whale in population_woa: Calculate_Whale_Parameters(A, C, l, p) IF p < 0.5 THEN IF A < 1 THEN Apply_Shrinking_Encircling(whale, best_solution_woa) ELSE Apply_Search_For_Prey(whale, random_whale) END IF ELSE Apply_Spiral_Update(whale, best_solution_woa, l) END IF Evaluate_Whale_Fitness() END FOR Update_Best_Solution_WOA() END FOR // Phase 3: Arithmetic Optimization Algorithm (Fine-tuning) Initialize_Agent_Population(population_aoa, 20, best_solution_woa) best_solution_aoa = best_solution_woa $\alpha = 0.1$; $\mu = 0.499$ FOR iteration = 1 to max_iterations_aoa DO MOP = 1 - ((iteration / max_iterations_aoa) $^{\alpha}$) FOR each agent in population_aoa: r1 = random() IF r1 > MOP THEN // Exploration phase r3 = random() IF r3 < 0.5 THEN Apply_Addition_Operator(agent, best_solution_aoa, MOP, μ) ELSE Apply_Subtraction_Operator(agent, best_solution_aoa, MOP, μ) END IF ELSE // Exploitation phase r2 = random() IF r2 < 0.5 THEN Apply_Multiplication_Operator(agent, best_solution_aoa, MOP, μ) ELSE Apply_Division_Operator(agent, best_solution_aoa, MOP, μ) END IF END IF Evaluate_Agent_Fitness() END FOR Update_Best_Solution_AOA() END FOR RETURN best_solution_aoa END GWO-WOA-AOA </pre>	<pre> FUNCTION Initialize_Wolf_Population(size): FOR i = 1 to size: FOR d = 1 to dimension: population[i, d] = random(lower_bound, upper_bound) END FOR END FOR RETURN population FUNCTION Update_Leadership_Hierarchy(): Sort_Population_By_Fitness() α = Best_Solution() β = Second_Best_Solution() δ = Third_Best_Solution() FUNCTION Calculate_A_and_C_Vectors(): A1 = a * (2 * random(dimension) - 1) A2 = a * (2 * random(dimension) - 1) A3 = a * (2 * random(dimension) - 1) C1 = 2 * random(dimension) C2 = 2 * random(dimension) C3 = 2 * random(dimension) RETURN A1, A2, A3, C1, C2, C3 FUNCTION Update_Wolf_Position(wolf, α, β, δ, a): D α = C1 * α - wolf D β = C2 * β - wolf D δ = C3 * δ - wolf X1 = α - A1 * D α X2 = β - A2 * D β X3 = δ - A3 * D δ wolf_new = clip((X1 + X2 + X3) / 3, lower_bound, upper_bound) RETURN wolf_new FUNCTION Calculate_Whale_Parameters(A, C, l, p): A = 2 * a * random(dimension) - a C = 2 * random(dimension) l = (a - 1) * random() + 1 p = random() RETURN A, C, l, p FUNCTION Apply_Shrinking_Encircling(whale, best_whale): D = C * best_whale - whale whale_new = clip(best_whale - A * D, lower_bound, upper_bound) RETURN whale_new FUNCTION Apply_Search_For_Prey(whale, random_whale): D = C * random_whale - whale whale_new = clip(random_whale - A * D, lower_bound, upper_bound) RETURN whale_new FUNCTION Apply_Spiral_Update(whale, best_whale, l): D = best_whale - whale whale_new = clip(D * exp(l) * cos(2π * l) + best_whale, lower_bound, upper_bound) RETURN whale_new FUNCTION Apply_Addition_Operator(agent, best_solution, MOP, μ): agent_new = clip(best_solution / (MOP + ϵ) * ((upper_bound - lower_bound) * μ + lower_bound), lower_bound, upper_bound) RETURN agent_new FUNCTION Apply_Subtraction_Operator(agent, best_solution, MOP, μ): agent_new = clip(best_solution * MOP * ((upper_bound - lower_bound) * μ + lower_bound), lower_bound, upper_bound) RETURN agent_new FUNCTION Apply_Multiplication_Operator(agent, best_solution, MOP, μ): agent_new = clip(best_solution - MOP * ((upper_bound - lower_bound) * μ + lower_bound), lower_bound, upper_bound) RETURN agent_new FUNCTION Apply_Division_Operator(agent, best_solution, MOP, μ): agent_new = clip(best_solution + MOP * ((upper_bound - lower_bound) * μ + lower_bound), lower_bound, upper_bound) RETURN agent_new </pre>

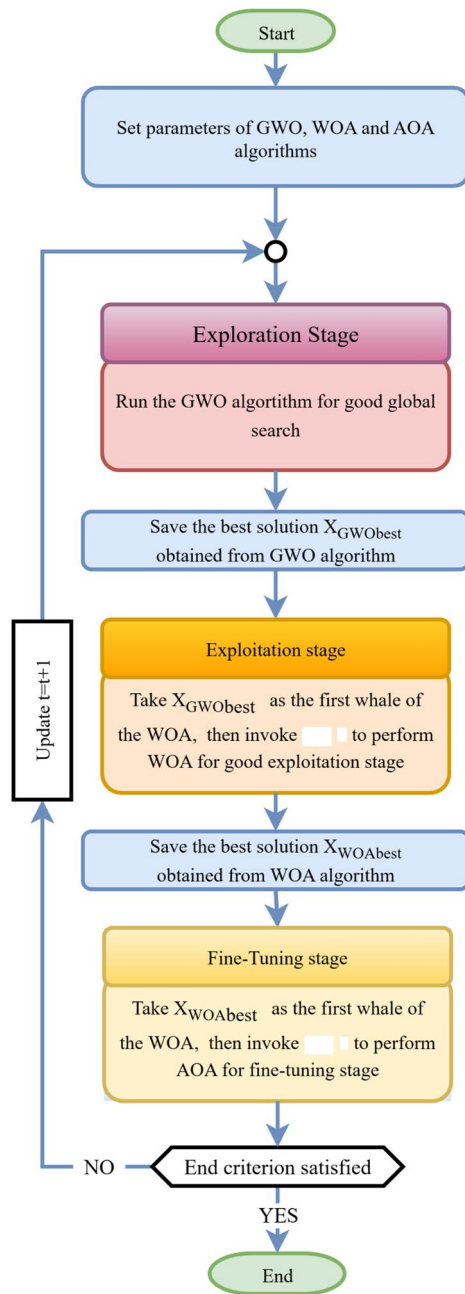


Fig. 6 Flowchart of hybrid GWO-WOA-AOA method

The operating principle of hybrid GWO-WOA-AOA technique is shown in detail in the flow diagram in Fig. 6 and Pseudocode. All stages and general functioning of this technique are summarized below:

The algorithm begins with the defining the parameters (size, limits, purpose function) of the optimization problem and the number of iterations for each sub -algorithm and the number of iterations. For all sub -algorithms, the population size is 20 and the iteration number is 100.

Table 2 CEC 2019 function properties

No.	Function	D	Search Range	B
$\mathcal{F}_1$	Storn's Chebyshev Polynomial Fitting Problem	9	$[-8192, 8192]$	1
$\mathcal{F}_2$	Lennard-Jones Minimum Energy Cluster	18	$[-4, 4]$	1
$\mathcal{F}_3$	Inverse Hilbert Matrix Problem	16	$[-16,384, 16,384]$	1
$\mathcal{F}_4$	Weierstrass Function	10	$[-100, 100]$	1
$\mathcal{F}_5$	Ackley Function	10	$[-100, 100]$	1
$\mathcal{F}_6$	Griewangk's Function	10	$[-100, 100]$	1
$\mathcal{F}_7$	Happy Cat Function	10	$[-100, 100]$	1
$\mathcal{F}_8$	Rastrigin's Function	10	$[-100, 100]$	1
$\mathcal{F}_9$	Expanded Schaffer's Function	10	$[-100, 100]$	1
$\mathcal{F}_{10}$	Modified Schwefel's Function	10	$[-100, 100]$	1

In the first stage, the GWO algorithm, which stands out with its exploration ability, comes into play. At this stage, the 20 -individual wolf population is initiated randomly and the suitability value of each solution is calculated. Solutions in the population are sorted according to their performances, alpha (best), beta (second best) and delta (third best) leading wolves are identified. During 100 iteration, each iteration is first updated by parameter A and this parameter is reduced from 2 to 0. Then A and C vectors are calculated for each wolf. Omega wolves (ordinary individuals) update their positions according to the positions of Alpha, Beta and Delta leaders. The best solution in the population is recorded and updated in each iteration. When the GWO phase is completed, the best solution is transferred to the next stage as the starting point.

In the second stage, exploitation process is performed with the WOA algorithm. The whale population is started with 20 individuals around the best solution from GWO. At this stage, a more intense and focused search is done around the promising zone GWO has found. In each iteration, parameter A is updated and the parameters A, C, L and P, which control whale behavior, are calculated. For each whale, two different motion strategy is applied according to the P value produced randomly: If $P < 0.5$, the circle mechanism ($|A| < 1$) or prey search behavior ($|A| \geq 1$) is used according to the size of the vector A. If $P \geq 0.5$, the spiral update mechanism is activated. In each iteration, the suitability values of whales are calculated and the best solution is updated. At the end of 100 iterations, the best solution that WOA has found is transferred to the final stage.

In the third and final stage, fine tuning is made with the AOA algorithm. The agent population is started with 20 individuals around the best solution from WoA. At this stage, the alpha (α) value of the algorithm is selected as 0.1 and Mu (μ) to increase the fine talent of the algorithm. In each iteration, the ratio of mathematical operator (mop) is calculated first and this value decreases as the iteration

Table 3 Control parameters employed in optimization approaches

Algorithm	Parameter	Value
GWO_WOA_AOA	GWO	ps, mi, a, A, C
	WOA	ps, mi, a, b, A, C, p
	AOA	ps, mi, α , μ , r_1 , r_2 , r_3
GWO	ps, mi, a, A, C	20, 300, 2 $\rightarrow$ 0, $[-a, a]$, $[0, 2]$
WOA	ps, mi, a, A, C, P	20, 300, 2 $\rightarrow$ 0, $[-a, a]$, $[0, 2]$, $[0, 1]$
AOA	ps, mi, α , μ , r_1 , r_2 , r_3	20, 300, 0.1, 0.499, $[0, 1]$, $[0, 1]$, $[0, 1]$
ALO	ps, mi, eae	20, 300, 0.1
PSO	ps, mi, w, c_1 , c_2	20, 300, 0.5, 1.5, 1.5
DE	ps, mi, F , C_R	20, 300, 0.5, 0.9
RIME	ps, mi, step0, cool, dep_prob	20, 300, 0.25 $\times$ (ub-lb), 1-t/(mi+1), 0.4+0.4 $\times$ cool

Table 4 Performance comparison of the suggested hybrid GWO-WOA-AOA with GWO, WOA, AOA, RIME, ALO, PSO and DE

Function	Metrics	DE	PSO	ALO	RIME	AOA	GWO	WOA	GWO_WOA_AOA
Ackley	Best	2.145e+01	2.115e+01	2.132e+01	2.139e+01	2.134e+01	2.140e+01	2.130e+01	2.126e+01
	Mean	2.166e+01	2.157e+01	2.168e+01	2.167e+01	2.183e+01	2.167e+01	2.157e+01	2.156e+01
	Std	1.100e-01	1.361e-01	1.159e-01	1.000e-01	1.938e-01	1.084e-01	1.220e-01	1.190e-01
Chebyshev	Best	5.552e+04	3.677e+05	1.538e+00	3.042e+04	1.015e+00	6.317e+02	1.011e+00	1.000e+00
	Mean	2.667e+06	1.118e+07	1.696e+00	1.082e+06	4.125e+07	1.016e+06	4.082e+07	1.000e+00
	Std	4.095e+06	3.251e+07	7.882e-02	1.712e+06	7.535e+07	1.263e+06	7.236e+07	0.000e+00
Escaffer6	Best	4.157e+00	4.128e+00	5.197e+00	3.962e+00	4.769e+00	3.522e+00	4.094e+00	3.452e+00
	Mean	4.798e+00	4.961e+00	5.530e+00	4.558e+00	5.385e+00	4.552e+00	4.970e+00	4.467e+00
	Std	2.253e-01	3.190e-01	1.196e-01	2.637e-01	1.968e-01	3.453e-01	3.113e-01	1.189e-01
Griewank	Best	2.522e+00	7.756e+03	8.344e+05	3.157e+01	1.272e+06	3.916e+02	2.316e+04	3.173e+01
	Mean	2.099e+04	1.580e+05	1.372e+06	5.945e+01	2.962e+06	9.287e+03	4.405e+05	2.837e+03
	Std	5.035e+04	1.491e+05	2.863e+05	1.578e+01	8.552e+05	1.393e+04	3.789e+05	6.334e+03
HappyCat	Best	1.460e+00	3.050e+01	1.128e+04	2.624e+00	8.731e+03	2.825e+00	8.751e+02	3.250e+00
	Mean	5.966e+01	8.810e+02	2.157e+04	3.455e+00	3.055e+04	1.366e+02	6.287e+03	6.593e+01
	Std	1.154e+02	1.181e+03	4.092e+03	3.718e-01	9.287e+03	2.090e+02	3.310e+03	7.032e+01
Hilbert	Best	4.475e+00	5.183e+00	5.257e+00	4.436e+00	5.000e+00	4.463e+00	5.001e+00	4.397e+00
	Mean	5.319e+00	7.610e+00	5.295e+00	5.196e+00	5.000e+00	5.436e+00	9.846e+00	4.850e+00
	Std	6.262e-01	1.818e+00	1.864e-02	6.896e-01	0.000e+00	5.541e-01	4.402e+00	1.766e-01
Lennard-Jones	Best	8.669e+00	4.370e+00	4.751e+00	3.436e+00	1.156e+01	1.706e+00	4.285e+00	1.035e+00
	Mean	1.023e+01	8.755e+00	9.696e+00	6.400e+00	1.240e+01	5.413e+00	7.955e+00	3.237e+00
	Std	7.967e-01	1.896e+00	1.707e+00	1.329e+00	3.119e-01	2.194e+00	1.687e+00	1.464e+00
Rastrigin	Best	5.426e+01	1.033e+03	2.083e+05	4.792e+01	1.037e+05	9.961e+01	1.763e+04	8.778e+01
	Mean	1.994e+03	2.188e+04	4.755e+05	8.897e+01	5.595e+05	1.530e+03	1.469e+05	1.054e+03
	Std	6.315e+03	2.618e+04	1.238e+05	1.156e+01	1.716e+05	1.397e+03	7.629e+04	1.395e+03
Schwefel	Best	2.187e+03	1.833e+03	3.992e+04	2.268e+03	1.533e+04	2.043e+03	7.030e+03	1.467e+03
	Mean	2.957e+03	7.204e+03	9.280e+04	2.836e+03	4.351e+04	3.177e+03	1.852e+04	2.632e+03
	Std	6.018e+02	4.987e+03	2.078e+04	2.465e+02	5.847e+03	5.379e+02	6.617e+03	4.262e+02
Weierstrass	Best	1.894e+02	1.860e+02	1.897e+02	1.909e+02	1.891e+02	1.889e+02	1.881e+02	1.872e+02
	Mean	1.924e+02	1.897e+02	1.940e+02	1.923e+02	1.934e+02	1.915e+02	1.910e+02	1.901e+02
	Std	8.036e-01	1.448e+00	1.369e+00	6.484e-01	1.174e+00	1.170e+00	1.301e+00	1.243e+00

progresses. For each agent, a random R1 value is produced and this value is compared with the mop. If the $R1 > mop$ is the discovery phase is active and the collection or subtraction operator is applied according to random R3 value. If the $R1 \leq mop$ is active, the exploitation phase is active and a multiplication or division operator is applied according to random R2 value. In each iteration, the compliance values of agents are calculated and the best solution is updated. At the end of 100 iteration, AOA's best solution is returned as the ultimate optimal solution of the hybrid algorithm.

The hybrid GWO-WOA-AOA algorithm can use the powerful aspects of each sub-algorithm in order to scan the solution space comprehensively and can make effective exploitation and fine-tuning on the solutions. The GWO makes a large -scale discovery, WOA concentrates in promising areas, and AOA maximizes the sensitivity and accuracy of the solution by making the latest touches. Thanks to these three -stage structure, more balanced, faster and higher quality results can be achieved than the classic algorithms used alone. Throughout the algorithm, all solutions

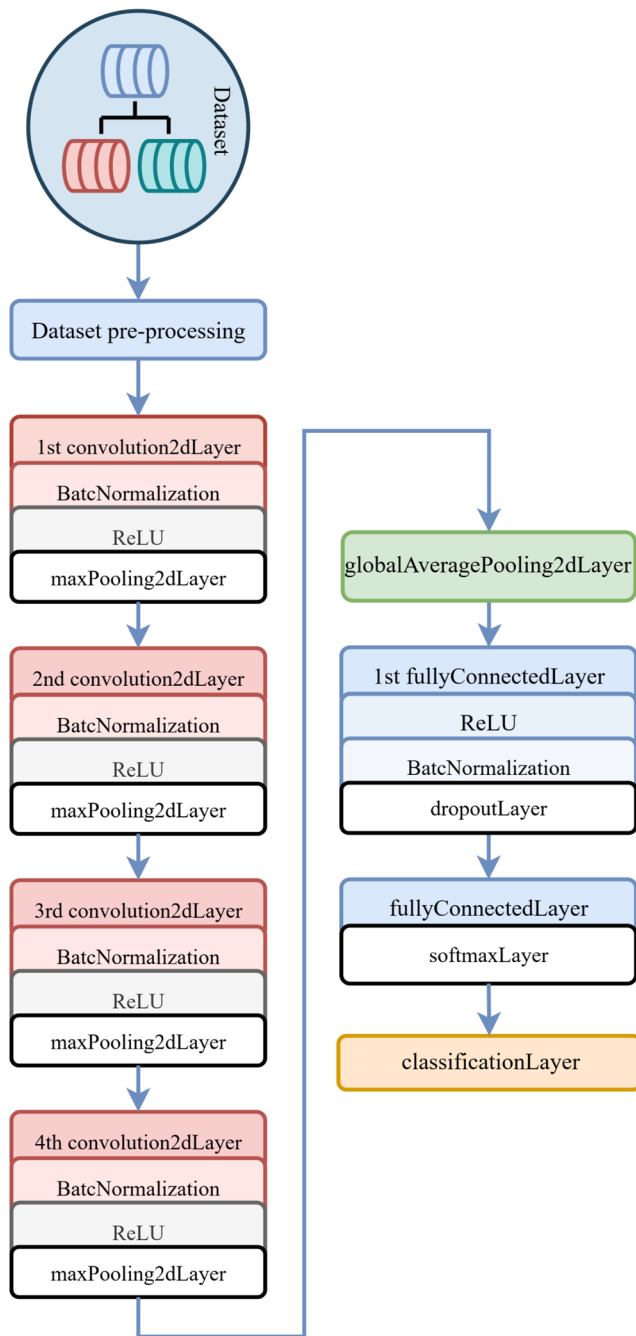


Fig. 7 Proposed of 2D-CNN structure

are kept within the defined boundaries and the best solution at each stage is transferred to the next stage and the loss of information is prevented.

The performance evaluation of the suggested GWO-WOA-AOA approach is performed on the CEC 2019 benchmark functions. Table 2 contains the CEC 2019 functions used. These benchmark functions were proposed within the scope of IEEE Congress on Evolutionary Computation 2019 and provide a reliable and general testbed for optimization algorithms [39]. Thanks to the complex function portfolio

Table 5 Search ranges for optimizing hyperparameters of CNN model by hybrid GWO-WOA-AOA method

Hyperparameter	Search Range	Type
Number of filters in Conv2d layers	[8, 16, 32, 64], [16, 32, 64, 128], [32, 64, 128, 256], [64, 128, 256, 512]	Discrete
Kernel size	[3, 5, 7, 9, 11]	Discrete
Pooling size	[2, 3, 4, 5, 6]	Discrete
Number of neurons in the 1st fully connected layer	[16, 32, 64, 128, 256]	Discrete
Dropout rate	(0.1, 0.5)	Continuous
Learning rate	(0.0001, 0.01)	Continuous
Minibatch size	[8, 16, 32, 64, 100, 128]	Discrete
Optimizer	['adam', 'sgd', 'rmsprop']	Categorical
Max epoch	(10, 100)	Continuous

included, the ability of the algorithms to deal with non-separable and high-dimensional problems with multiple local minima can be evaluated fairly and comprehensively. In this way, the performances of different algorithms become comparable with each other.

In Table 3, the control parameters of the GWO, WOA and AOA algorithms used in the proposed hybrid optimization method and the basic parameter values of the selected Ant Lion Optimization (ALO), Differential Evolution (DE), PSO and Rime-Inspired Metaheuristic (RIME) [40] techniques for comparison are presented in detail. In order to conduct the comparison fairly, the population size ($ps=20$) and the number of running cycles are kept constant as 50 in all algorithms. In the hybrid method, the maximum iteration number of each algorithm is 100, and a total of 300 iterations are performed. In order to evaluate the other methods under the same conditions, 300 iterations are performed for these algorithms and the parameter values determining the unique search strategy of each algorithm are selected from the ranges commonly used in the literature.

In Table 4, the performance of the suggested GWO-WOA-AOA hybrid approach on the CEC 2019 test functions, together with the basic methods (GWO, WOA and AOA) that constitute the algorithm, is presented in comparison with common optimization algorithms such as RIME, ECO, ALO, PSO and DE. All algorithms are assessed with same the number of iterations and population size, thus providing a fair comparison environment. The findings show that the suggested hybrid technique provides a notable improvement in the context of both best and mean results, especially compared to the results obtained by GWO, WOA and AOA alone. For example, the mean and best values of the hybrid approach for the Chebyshev, Escaffer6, Lennard-Jones and Weierstrass functions are very close to the ideal result of 1 or directly 1. In addition, the low standard deviation (Std) values show that the algorithm produces stable and reliable results. The GWO-WOA-AOA algorithm has

yielded more successful results not only compared to its own components, but also compared to other popular methods such as RIME, ECO, ALO, PSO and DE. These results show that the hybrid structure provides a more effective balance of exploration, exploitation and fine tuning by combining the strengths of GWO, WOA and AOA, and offers superior performance in complex optimization problems. As a result, the proposed hybrid algorithm offers a competitive and reliable solution compared to both classical and modern optimization techniques.

3.3 Proposed CNN algorithm for heart disease prediction

In this section, the general structure, layers and operation of the 2D-CNN architecture developed for heart disease prediction are detailed. It is emphasized that the optimization of hyperparameters (number of filters, kernel size, pooling size, mini-batch size, number of neurons in the fully connected layer, dropout rate, optimizer type, learning rate, and max epoch) that directly affect the success of the model is performed with the proposed hybrid GWO-WOA-AOA algorithm. In addition, the problem definition of hyperparameter optimization, the search ranges used and the fitness function are explained.

3.3.1 2D-CNN model architecture overview

The architecture of the 2D-CNN model created for heart attack risk prediction is given in Fig. 7. The model's input layer receives a dataset containing 11 features associated with heart disease risk in a two-dimensional matrix format (height, width, number of channels). The data is preprocessed with z-score normalization and missing data removal before being fed into the model.

The detailed structural features of the proposed CNN architecture in Fig. 7 are as follows: The model is dynamically built using hyperparameters optimized within the ranges specified in Table 5. The architecture has four consecutive Conv2D layers, and the number of filters in these layers is determined by one of the four different configuration options: [8, 16, 32, 64], [16, 32, 64, 128], [32, 64, 128, 256] or [64, 128, 256, 512]. Each of these configurations represents the number of filters in the first, second, third, and fourth Conv2D layers, respectively. For example, when the [32, 64, 128, 256] configuration is selected, the first Conv2D layer contains 32 filters, the second layer contains 64 filters, the third layer contains 128 filters, and the fourth layer contains 256 filters. All convolutional layers use the same kernel size, selected from the range [3, 5, 7, 9, 11] specified in Table 5. Each Conv2D layer extracts increasingly complex spatial features from the data by applying

learnable filters to the input. Using different kernel sizes allows for the detection of patterns at multiple scales and increases the model's ability to capture a wider range of features. This structure allows for the capture of spatial relationships between features. After each convolutional layer, batch normalization is applied, followed by the ReLU activation function. Batch normalization speeds up the training process and ensures more stable model learning. ReLU activation, by setting negative values equal to zero, enables the model to learn nonlinear relationships and prevents the vanishing gradient problem.

Following each Conv2D layer, MaxPooling2D layers are applied with pooling sizes selected from the range [2–6]. All pooling layers use the same size, and these layers reduce the spatial dimension of the feature maps, preserving important features while reducing computational cost and helping to prevent model overfitting.

The global average pooling layer converts the feature maps from the last convolution block into a single vector. This process summarizes the spatial information by averaging each feature map and significantly reduces the number of parameters, improving overall performance and enhancing the model's generalization ability.

The first fully connected layer has a number of neurons selected from the range [16, 32, 64, 128, 256], and ReLU activation is applied. The number of neurons in this layer determines the model's ability to integrate high-level features and serves as an important hyperparameter to be optimized. This layer transforms the features from global average pooling into high-level abstract representations.

The dropout layer is configured with a dropout rate selected from the continuous range of 0.1–0.5. The dropout rate controls the proportion of neurons deactivated in each training iteration. By deactivating random neurons during training, dropout prevents the model from becoming overly dependent on specific features and improves generalization performance.

The second fully connected layer contains neurons equal to the number of classes (2), and a softmax activation function is used in the output layer. The softmax function generates probability values between 0 and 1 for each class, and the sum of the probabilities of all classes equals 1. This allows the model to predict the probability that each individual belongs to a heart disease risk class.

Categorical cross-entropy is used as the loss function in model training. This loss function measures the difference between the predicted probability distribution and the true class labels and is the most commonly used loss function for binary classification problems. The optimization algorithm is selected from the options ['adam', 'sgd', 'rmsprop']. The Adam optimizer applies different update steps for each parameter using an adaptive learning rate and achieves

rapid convergence through the momentum mechanism. SGD (Stochastic Gradient Descent) uses the classical gradient descent method, while RMSprop adaptively adjusts the learning rate. The default momentum parameters for the Adam optimizer are $\beta_1=0.9$, $\beta_2=0.999$, and $\varepsilon=1e-7$.

The learning rate is optimized in the continuous range of 0.0001–0.01.0001.01. Because the learning rate determines the amount of change in the parameters with each update, it plays a direct role in how quickly the model converges. Too high a learning rate can cause the model to be unstable, while too low a learning rate can increase the training time.

The mini-batch size is determined from the options [8, 16, 32, 64, 100, 128]. The batch size determines the number of samples used in each iteration during training and helps balance computational speed with the quality of the learned parameters. Larger batch sizes provide more stable gradient estimates, while smaller batch sizes can increase the model's generalization ability.

The maximum epoch count is optimized in the continuous range of 10–100. The epoch count specifies the number of times the entire dataset is exposed to the model and controls the degree to which the model has learned. During model training, an early-stop mechanism is implemented that terminates training if there is no improvement in the validation loss over five epochs. This mechanism prevents the model from overfitting and ensures that the model weights with the best generalization performance are retained.

3.3.2 Hyperparameter optimization problem in 2D-CNN model training

In this study, heart attack risk prediction is performed using the 2D-CNN architecture shown in Fig. 7. For increasing the classification success of the CNN approach, the important hyperparameters of the approach are optimized with the

proposed hybrid GWO-WOA-AOA optimization method. The hyperparameters to be optimized by the proposed hybrid optimization are the number of filters in CNN layers, kernel size, pooling size, number of neurons in the 1st fully connected layer, dropout rate, learning rate, mini-batch size, type of optimizer and max epoch. Table 5 shows search range of the hyperparameters to be optimized.

The fitness function for the GWO-WOA-AOA algorithm is designed to optimize the hyperparameters of the CNN model to maximize the classification accuracy on the validation dataset. The objective function is formulated as a minimization problem and is defined as follows:

$$f(x) = 1 - (TP + TN)/(TP + TN + FP + FN) \quad (23)$$

In Eq. (23), $f(x)$ represents the fitness value to be minimized; FN, FP, TN, and TP represent the number of false negative, false positive, true negative, and true positive samples on the validation data set, respectively.

Hyperparameter vector, $x = [x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9]$, represents the variables “Number of filters in Conv2D layers”, “Kernel size”, “Pooling size”, “Number of neurons in the 1st fully connected layer”, “Dropout rate”, “Learning rate”, “Minibatch size”, “Optimizer” and “Maxepochs” specified in Table 5, respectively.

The fitness function is designed as a minimization problem, and low fitness values indicate superior model performance. When perfect classification accuracy is achieved, the optimal solution approaches zero, which means that the validation accuracy rate is 100%. Thanks to this formulation, the GWO-WOA-AOA hybrid optimization algorithm systematically explores the hyperparameter search space, enabling the CNN model to achieve the highest accuracy in heart attack risk prediction on the validation dataset.

3.3.3 Optimizing hyperparameters using proposed hybrid optimization algorithm

In this study, hyperparameter optimization of the suggested 2D-CNN approach for heart attack risk prediction was performed using the hybrid GWO-WOA-AOA algorithm. The performance of the suggested hybrid approach was compared with that of the GWO, WOA, AOA, and RIME approaches used alone.

The dataset used in the optimization process was divided into three parts so as to evaluate the performance of the approach fairly and reliably: 70% of the data was reserved for training, 10% for validation, and 20% for testing. Only training and validation datasets were used in hyperparameter optimization, and the performance of the final approach was measured on the reserved test dataset.

Table 6 Hyperparameter results of the suggested hybrid and other approaches

Hyperparameter	RIME	AOA	GWO	WOA	GWO-WOA-AOA
Number of filters	[32, 64, 128, 256]	[8, 16, 32, 64]	[16, 32, 64, 128]	[8, 16, 32, 64]	[32, 64, 128, 256]
Kernel size	9	5	3	5	11
Pooling size	3	2	2	3	3
Number of neurons in 1st fclayer	128	16	64	32	128
Dropout rate	0.294	0.194	0.173	0.247	0.313
Learning rate	0.00023	0.00045	0.00032	0.00059	0.00015
Mini-batch size	100	64	100	32	128
Optimizer	Adam	Adam	Adam	Adam	Adam
Maxepoch	44	77	70	82	36
Computation Time (min)	30.8	25.4	26.9	28.3	35.6

In the hybrid GWO-WOA-AOA algorithm, each sub-algorithm was run for 10 iterations respectively, completing 30 iterations in total. For comparison purposes, GWO, WOA, AOA and RIME algorithms were run independently for 30 iterations. All algorithms optimized nine basic hyperparameters of the CNN model within the ranges given in Table 5: the number of filters in Conv2D layers, kernel size, pooling size, number of neurons in the first fully connected layer, dropout rate, learning rate, mini-batch size, optimizer type and max epoch. Hyperparameters are optimized according to the generated fitness function. This function is designed to maximize the classification performance on the validation dataset. Thus, lower fitness value means higher accuracy.

Table 6 presents best hyperparameter combinations obtained for each algorithm. According to the Table 6, the hybrid GWO-WOA-AOA algorithm obtained a more balanced and effective hyperparameter set compared to the other methods and provided the highest classification performance on the validation dataset. These results reveal that the hybrid algorithm scans the search space more comprehensively and efficiently, allowing the model to perform better regarding both generalization and accuracy.

Table 6 presents the total computational time in minutes required for hyperparameter optimization for each optimization algorithm. All algorithms were run on the same hardware infrastructure (Intel Core i7-6500U @ 2.50 GHz, 16 GB RAM). The proposed GWO-WOA-AOA method required approximately 35.6 min of computation for 30 iterations. This time is due to the three-stage sequential optimization structure (GWO: 10 iterations, WOA: 10 iterations, AOA: 10 iterations). For comparison, the RIME algorithm took 30.8 min, AOA 25.4 min, GWO 26.9 min, and WOA 28.3 min. The approximately 20–40% longer time

required for the suggested hybrid approach compared to the other approaches is due to the sequential execution of the three algorithms and the training of the CNN model at each stage. However, considering the superior performance achieved (96.22% accuracy), this additional computational cost is acceptable and does not pose a significant obstacle to practical use in clinical applications, as it is a one-time optimization process. Each iteration involves training the CNN model, hyperparameter evaluation, and fitness calculation. On average, each iteration takes approximately 1.2 min.

Furthermore, the convergence graph of the suggested hybrid method and the comparative methods according to the fitness function are presented in Fig. 8. As indicated in Fig. 8, the hybrid GWO-WOA-AOA algorithm showed more stable convergence than the other methods. This enabled the model to reach the optimum hyperparameter combination in more reliably.

Based on the convergence curve in Fig. 8, the hyperparameters obtained with the hybrid GWO-WOA-AOA algorithm maximized the performance of the approach on the validation dataset and outperformed the results obtained with classical methods. This shows that the suggested hybrid method yields an successful and reliable solution for hyperparameter optimization in complex classification problems such as heart attack risk prediction.

4 Results and discussion

A computer with an Intel(R) Core(TM) i7-6500U CPU @ 2.50 GHz and 16 GB RAM was used to perform all experiments in this paper. MATLAB and Python programming languages were used in data processing, model training and hyperparameter optimization processes. In the Python environment, necessary deep learning and machine learning libraries (e.g. TensorFlow/Keras, scikit-learn) were used, as well as MATLAB's optimization and data analysis tools. All algorithms and comparative experiments were conducted on this hardware and software infrastructure.

4.1 Performance metrics

In this study, commonly used performance metrics such as accuracy, F1 score, sensitivity/recall, precision, Negative Predictive Value (NPV), Matthews Correlation Coefficient (MCC) and Cohen's Kappa (κ) coefficient were employed to assess the success of the developed heart disease prediction model. These metrics are of great importance in terms of revealing how successful the model is in different aspects.

Accuracy: It represents the percentage of examples that the model classifies correctly among all examples. It shows how accurate the model's predictions are in general.

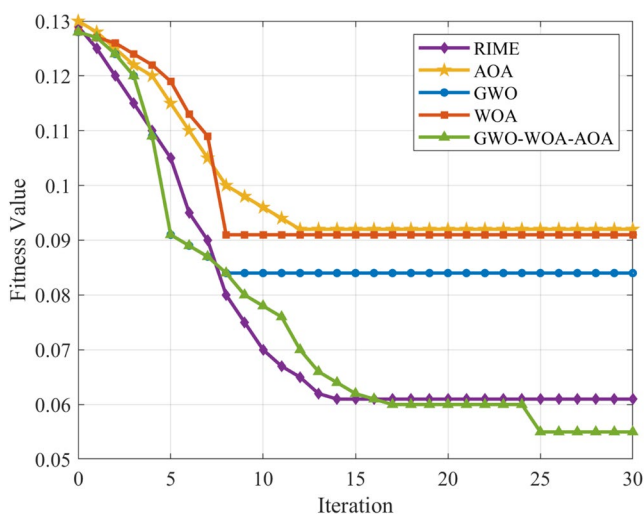
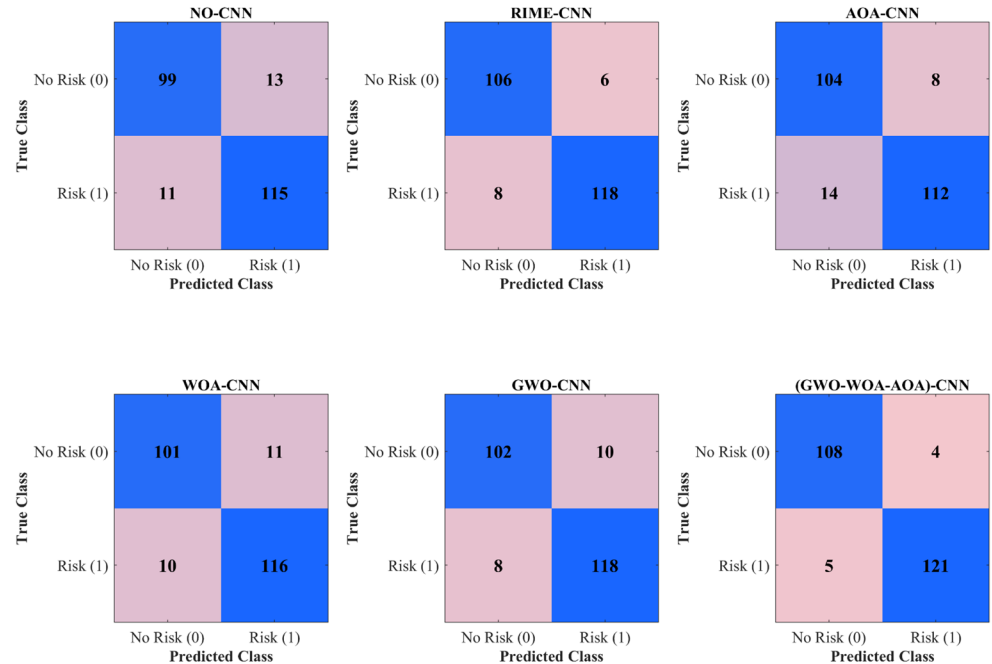


Fig. 8 Convergence curves of GWO-WOA-AOA algorithm and other algorithms

Table 7 Comparison of classification success of different methods on heart attack data

Method	Accuracy (%)	F1_Score (%)	Recall (%)	Precision (%)	NPV (%)	MCC (%)	Kappa (%)
NO-CNN	89.92	90.55	91.27	89.84	90.00	79.75	79.74
RIME-CNN	94.12	94.40	93.65	95.16	92.98	88.22	88.21
AOA-CNN	90.76	91.06	88.89	93.33	88.14	81.61	81.50
WOA-CNN	91.18	91.70	92.06	91.34	90.99	82.29	82.28
GWO-CNN	92.44	92.91	93.65	92.19	92.73	84.82	84.81
GWO-WOA-AOA-CNN	96.22	96.41	96.03	96.80	95.58	92.42	92.41

Fig. 9 Confusion matrices of heart attack risk prediction results of models

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (24)$$

$$Precision = \frac{TP}{TP + FP} \quad (27)$$

In Eq. 24, FN is the number of False Negative samples; FP is the number of False Positive samples; TN is the number of True Negative samples; TP is the number of True Positive samples.

F1 score: It is the harmonic mean of precision and sensitivity values. It reflects the performance of the model more accurately, especially in unbalanced data sets.

$$F1_{score} = 2 * \frac{Precision * Sensitivity}{Precision + Sensitivity} \quad (25)$$

Sensitivity/Recall: It shows how many of the samples that should have been classified as positive were correctly detected as positive by the model.

$$Sensitivity = \frac{TP}{TP + FN} \quad (26)$$

Precision: It indicates how many of the samples predicted by the model as positive are actually positive.

Negative Predictive Value (NPV) measures the proportion of samples predicted as negative that are actually negative. In medical diagnostic systems, it is critical for assessing the reliability of disease absence predictions.

$$NPV = \frac{TN}{TN + FN} \quad (28)$$

Matthews Correlation Coefficient (MCC) is one of the most reliable performance metrics for binary classification problems. It provides objective results even on imbalanced datasets and takes all confusion matrix elements into account.

$$MCC = \frac{(TP * TN) - (FP * FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (29)$$

Cohen's Kappa (κ) coefficient is a metric that evaluates observer agreement and classification performance. It measures true classification success by taking random chance into account.

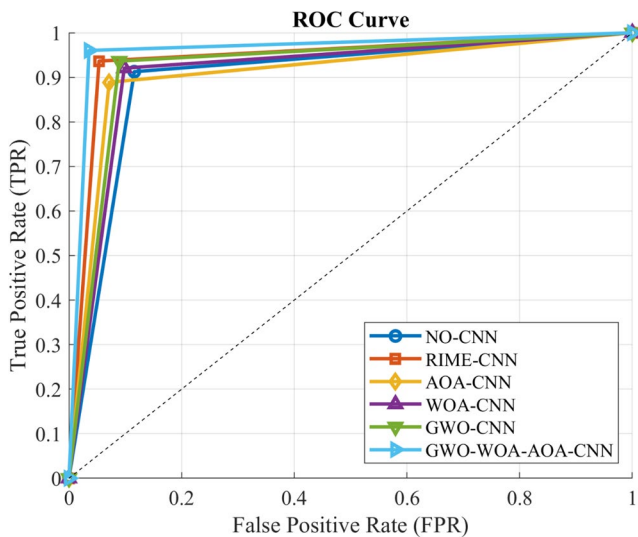


Fig. 10 ROC curves of GWO-WOA-AOA-CNN model and other models

$$\kappa = \frac{P_o - P_e}{1 - P_e}. \quad (30)$$

In Eq. 30, $P_o = \frac{TP+TN}{TP+TN+FP+FN}$,
 $P_e = \frac{(TP+FN)(TP+FP)+(FP+TN)(FN+TN)}{(TP+TN+FP+FN)^2}$.

These metrics were employed to assess the overall performance of the model and in particular how well it detects the positive class, thus providing a detailed overview of both overall and class-specific performance.

4.2 Evaluation in terms of performance metrics

In this study, the prediction of heart disease success of the developed methods was evaluated with common performance metrics such as accuracy, F1 score, sensitivity/recall, precision, NPV, MCC and Kappa. In the evaluations, 80% of the data set was separated as training and 20% as test data. The final performance measurements of the models were performed on the test data.

Table 7 presents the comparative results of the no optimization (NO)-CNN, RIME-CNN, AOA-CNN, WOA-CNN, GWO-CNN, and the proposed GWO-WOA-AOA-CNN methods. According to Table 7, the proposed

hybrid GWO-WOA-AOA optimized CNN model showed the highest performance with an accuracy rate of 96.22%. Similarly, the F1 score (96.41%), sensitivity (96.03%), precision (96.80%), NPV (95.58%), MCC (92.42%), and Kappa (92.41%) values are also higher compared to the other methods. These results show that the hybrid optimization method increases both the overall accuracy and the effectiveness of the model in distinguishing the positive class in complex datasets.

Figure 9 shows the confusion matrices for the NO-CNN, RIME-CNN, AOA-CNN, WOA-CNN, GWO-CNN, and the proposed GWO-WOA-AOA-CNN methods, respectively. Examining the confusion matrix of the proposed GWO-WOA-AOA-CNN approach in particular, it is seen that the number of misclassified samples is quite low, and the model can distinguish both positive (heart disease) and negative (no heart disease) classes with high accuracy. In the GWO-WOA-AOA-CNN model, 108 of the 112 patients whose true class was “No Risk (0)” were correctly predicted, while only 4 patients were incorrectly classified as “At Risk (1)” (False Positive - FP). Similarly, 121 of 126 patients whose true class was “Risk (1)” were correctly predicted, while only 5 patients were misclassified as “No Risk (0)” (False Negative - FN). These results demonstrate that the model can distinguish both classes with high precision. Examining the confusion matrices of the other models, it is seen that the NO-CNN model has 13 FP and 11 FN values, the RIME-CNN model has 6 FP and 8 FN values, the AOA-CNN model has 8 FP and 14 FN values, the WOA-CNN model has 11 FP and 10 FN values, and the GWO-CNN model has 10 FP and 8 FN values. This comparison reveals that the proposed GWO-WOA-AOA-CNN model provides the most balanced and reliable predictions by minimizing both FP and FN values. This reduction in the number of FPs and FNs demonstrates that the model minimizes the risk of misdiagnosis in clinical applications. Furthermore, the high precision value (96.80%) indicates that the vast majority of cases predicted as positive by the model actually have heart disease. The high sensitivity (96.03%) demonstrates that the model largely correctly identifies existing heart disease cases. The balanced and high F1 score and accuracy values demonstrate that the proposed approach provides reliable and stable results even with imbalanced datasets.

Table 8 Statistical significance analysis of classification methods on heart attack data (CV=5)

Method	Accuracy (%)	95% CI (Accuracy)	p-value (Accuracy)	F1_Score (%)	95% CI (F1_Score)	p-value (F1_Score)
NO-CNN	89.68	[88.12, 91.24]	-	90.31	[88.95, 91.67]	-
RIME-CNN	93.87	[92.45, 95.29]	0.028*	94.18	[92.84, 95.52]	0.031*
AOA-CNN	90.94	[89.38, 92.50]	0.084	91.22	[89.76, 92.68]	0.076
WOA-CNN	91.35	[89.91, 92.79]	0.063	91.88	[90.42, 93.34]	0.058
GWO-CNN	92.21	[90.85, 93.57]	0.042*	92.68	[91.34, 94.02]	0.039*
GWO-WOA-AOA-CNN	96.19	[94.72, 97.66]	0.016*	96.38	[94.95, 97.81]	0.014*

* $p < 0.05$ indicates statistically significant improvement over NO-CNN baseline

Fig. 11 Initial opening screen of the Heart Disease Prediction interface with blank input fields

Cardio Vascular Disease Prediction

Age		28-77	Resting ECG		0-2
Sex		0-1	Max Heart Rate		60-202
Chest Pain Type		1-4	Exercise Angina		0-1
Resting BP S		80-200	Oldpeak		0.0-6.2
Cholesterol		0-603	ST Slope		1-3
Fasting Blood Sugar		0-1			

Fig. 12 Interface screen showing “Heart disease risk: LOW.” result with low-risk data received from the user

Cardio Vascular Disease Prediction

Age	54	28-77	Resting ECG	0	0-2
Sex	0	0-1	Max Heart Rate	130	60-202
Chest Pain Type	2	1-4	Exercise Angina	0	0-1
Resting BP S	150	80-200	Oldpeak	0	0.0-6.2
Cholesterol	230	0-603	ST Slope	1	1-3
Fasting Blood Sugar	0	0-1			

Heart disease risk: LOW.

Figure 10 compares the ROC curves of the NO-CNN, RIME-CNN, AOA-CNN, WOA-CNN, GWO-CNN, and the proposed GWO-WOA-AOA-CNN models. The GWO-WOA-AOA-CNN model has a better discrimination ability than the other models, with a lower false positive rate (FPR) and a higher true positive rate (TPR). The ROC curve is located near the upper left corner, indicating a very high area under the curve (AUC). The ROC curves of all models lie well above the diagonal line (the random guess line), indicating that the models exhibit strong classification performance. However, the ROC curve of the GWO-WOA-AOA-CNN model is located above the curves of all other models, demonstrating the highest discrimination power. This demonstrates that the proposed hybrid optimization approach provides superior performance in terms of both sensitivity and specificity in heart disease risk prediction.

In conclusion, the classification success of the proposed hybrid approach (GWO-WOA-AOA-CNN) is remarkable. The model demonstrates high accuracy, F1 score, sensitivity, and precision, enabling it to be confidently used in clinical decision support systems. Confusion matrix and ROC curve analyses demonstrate that the model can distinguish both positive and negative classes with high accuracy, minimizing the risk of misdiagnosis.

To further verify the statistical significance of the performance improvements, Wilcoxon signed-rank tests were performed using 5-fold cross-validation. Table 8 presents the statistical analysis results, including 95% confidence intervals (CI) and p-values for accuracy and F1-score metrics, when NO-CNN was selected as the reference method. Statistical analysis results show that the RIME-CNN, GWO-CNN, and GWO-WOA-AOA-CNN methods

Fig. 13 Interface screen showing the “Heart disease risk: HIGH!” result with high-risk data received from the user

Cardio Vascular Disease Prediction

Age	65	28-77	Resting ECG	0	0-2
Sex	1	0-1	Max Heart Rate	87	60-202
Chest Pain Type	4	1-4	Exercise Angina	1	0-1
Resting BP S	140	80-200	Oldpeak	1.5	0.0-6.2
Cholesterol	306	0-603	ST Slope	2	1-3
Fasting Blood Sugar	1	0-1			

Heart disease risk: HIGH!

Table 9 Performance comparison with existing approaches

Methods	Data Used	Sample Size	F1-score (%)	Sensitivity (%)	Precision (%)	Accuracy (%)
MDCNN [41]	Cleveland	303	-	-	-	93.3
AIFS and MLP [15]	Statlog+Cleveland+Hungary+Switzerland+long beach VA	1190	87.21	88.5	86.05	87.7
PSO-based NN [11]	Hungary+Cleveland+Statlog	867	-	91.1	91.53	91.9
DHO based NN [11]	Hungary+Cleveland+Statlog	867	-	93.1	92.96	93.25
Hybrid GA and PSO+RF [12]	Statlog	270	-	95.56	-	91.4
SSAE+PSO [14]	Cleveland	303	95.8	98.8	-	96.1
EBPSO+KNN [16]	Cleveland	303	-	-	-	92.81
	Hungary	294	-	-	-	91.97
	Switzerland	123	-	-	-	91.62
EBPSO+SVM [16]	long beach VA	270	-	-	-	85.25
B-KHA+CapNet [18]	Cleveland	303	95	97	94	95
Ensemble model [24]	UCI Machine Learning dataset	303	95	95	97	96
Proposed study	HD dataset (long beach	1190	96.33	95.93	96.80	96.22
	VA+Switzerland+Hungary+Cleveland+Statlog)					

achieve statistically significant improvements over the reference NO-CNN approach ($p < 0.05$). Specifically, the proposed GWO-WOA-AOA-CNN method exhibited the most significant and statistically significant improvement, with $p = 0.016$ for accuracy and $p = 0.014$ for F1-score. The 95% confidence intervals further confirm the reliability and consistency of these improvements across different data partitions. While the AOA-CNN and WOA-CNN methods demonstrate numerical improvements, they fail to reach statistical significance ($p > 0.05$), suggesting that the performance gains may be due to random variation rather than systematic improvement. These statistical validation results confirm that the performance improvements achieved by the proposed hybrid optimization approach are not accidental but represent improvements in real predictive ability, and

support the clinical applicability and reliability of the developed model.

The Heart Disease Prediction interface developed in this study, examples of which are presented in Figs. 11, 12 and 13, provides early and automatic prediction of heart disease risk. Figure 11 includes a user-friendly interface where the user enters basic clinical parameters such as age, gender and chest pain type. After filling in the relevant fields, the user can click on the “Predict” button to get the model’s heart disease risk prediction. Figure 12 shows that when a low-risk sample data set is entered, the system displays the message “Heart disease risk: LOW.” in green. On the other hand, in Fig. 13, it is observed that the risk is high and the message “Heart disease risk: HIGH!” is highlighted in red as a result of the estimation made with a high-risk data set. This interface is an application example that can be used practically

in clinical decision support systems and contributes to early diagnosis processes, where the deep CNN model presented in the article, optimized with the hybrid GWO-WOA-AOA algorithm, runs in the background.

4.3 Comparison of proposed study with existing studies

The proposed GWO-WOA-AOA based CNN approach is rigorously evaluated through a comprehensive comparison with existing leading methods in the literature. Table 9 summarizes the results obtained by different deep learning and machine learning -based methods, organized by the data sets used and basic performance metrics such as the F1 score, sensitivity, precision and accuracy. Table 8 demonstrates that the suggested approach yields superior results in the context of accuracy (96.22%), F1 score (96.41%), sensitivity (96.03%), and precision (96.80%) when compared with most approaches in existing literature. In particular, the Modified Deep Convolutional Neural Network (MDCNN) [41] method, which only works on the Cleveland dataset, achieves 93.3% accuracy, while the proposed method achieves higher success on a larger and more diverse dataset created by merging five different datasets. Similarly, the Adaptive Infinite Feature Selection and Multi-Layer Perceptron (AIFS+MLP) [15] method lags behind the proposed method with 87.7% accuracy and 87.21% F1 score on the merged dataset.

When optimization-based methods are examined, approaches such as PSO-based neural network (PSO-NN) [11] and Deer Hunting Optimization-based neural network (DHO-NN) [11] generally produced results in the range of 91–93% accuracy. Hybrid Genetic Algorithm and PSO-optimized Random Forest (GA+PSO+RF) [12] stand out with 91.4% accuracy and 95.56% sensitivity on the Statlog dataset. However, most of these studies have been tested on either single or smaller datasets. The Stacked Sparse Auto-encoder and PSO (SSAE+PSO) [14] method, which uses deep learning and optimization together, has shown high performance with 96.1% accuracy and 95.8% F1 score on the Cleveland dataset. However, the success of this method has been evaluated in a more limited scope compared to the success of the proposed model on a large and heterogeneous dataset. Additionally, Enhanced Binary PSO with KNN (EBPSO+KNN) and SVM (EBPSO+SVM) based methods [16] provided accuracy rates between 85.25% and 92.81% on different datasets.

Recent studies using metaheuristic optimization and feature selection methods have also achieved remarkable results. The approach using Binary Krill Herd optimization (B-KHA) and the CapNet model proposed by Saran Kumar and Rekha [18] achieved 95% accuracy, 97% sensitivity,

94% precision, and 95% F1-score on the Cleveland dataset. Similarly, the ensemble approach developed by Arunachalam and Rekha [24], combining the k-Nearest Neighbor base classifier with X-boost, Adaboost, and Random Subspace classifiers, achieved 96% accuracy, 97% precision, 95% sensitivity, and 95% F-measure on the UCI Machine Learning dataset. However, these studies were tested on a single-source dataset and were not evaluated on heterogeneous data from different geographical regions. This constitutes a limitation in terms of the generalizability of the models. The proposed study, on the other hand, was tested on a comprehensive dataset created by merging five different datasets and demonstrated high performance on a wider and diverse range of patient profiles.

In the proposed study, successful results were obtained in all basic performance criteria compared to the approaches in the literature in the tests performed on one of the most comprehensive data sets in the literature, which was created by merging Long Beach VA, Switzerland, Hungary, Cleveland, and Statlog data sets. Especially the high F1 score and balanced sensitivity/precision ratios show that the model is strong in distinguishing both positive and negative classes. As a result, the findings in Table 9 clearly demonstrate that the proposed GWO-WOA-AOA optimized CNN model exhibits superior and balanced performance in heart disease prediction compared to existing approaches in the literature. The high success rates obtained on an extensive and heterogeneous dataset support the generalizability of the model and its reliable usability in clinical decision support systems.

4.4 Model explainability analysis

The explainability of model decisions is critical for the reliable use of artificial intelligence models in clinical decision support systems. In this paper, SHapley Additive Explanations (SHAP) analysis was performed to understand the feature importance and decision mechanisms in the prediction process of the proposed GWO-WOA-AOA optimized CNN model.

SHAP analysis uses a game-theoretic approach to calculate the contribution of each feature to model predictions, making the “black box” nature of the model transparent. Analysis results reveal which clinical parameters are most effective in predicting heart disease and help healthcare professionals understand model decisions.

The SHAP feature importance ranking, presented in Fig. 14, shows the most effective parameters in predicting heart disease. According to the results, ST slope stands out as the feature with the highest importance, with a mean absolute SHAP value of 0.138. This finding confirms that changes in the ST segment during exercise on

Fig. 14 SHAP feature importance ranking – Importance levels of features used in heart disease prediction ranked by their average absolute SHAP values

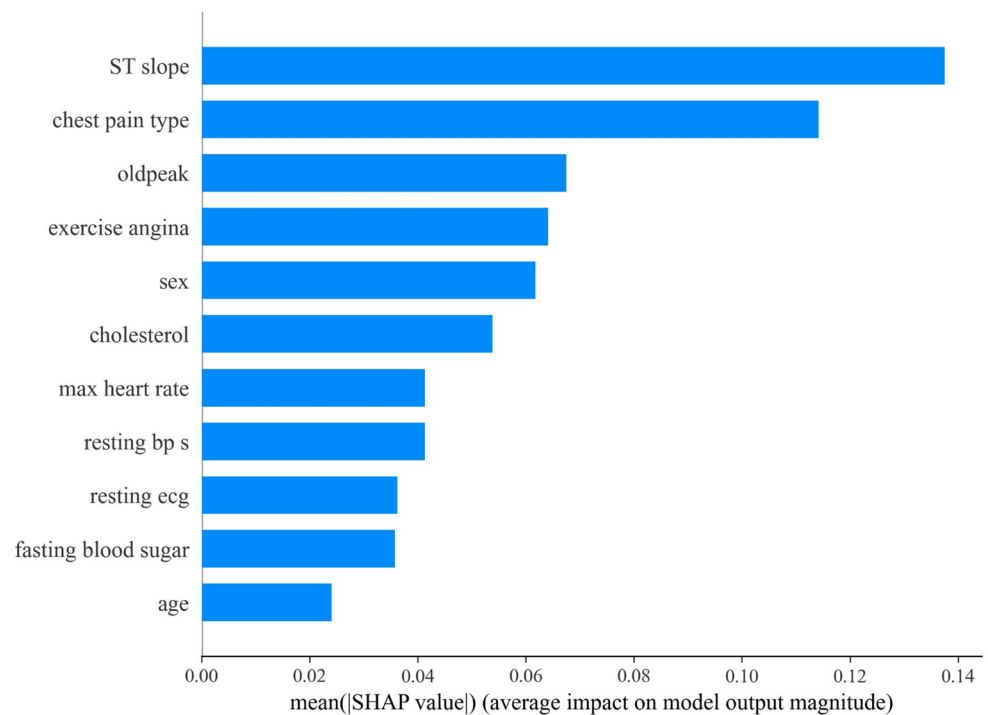
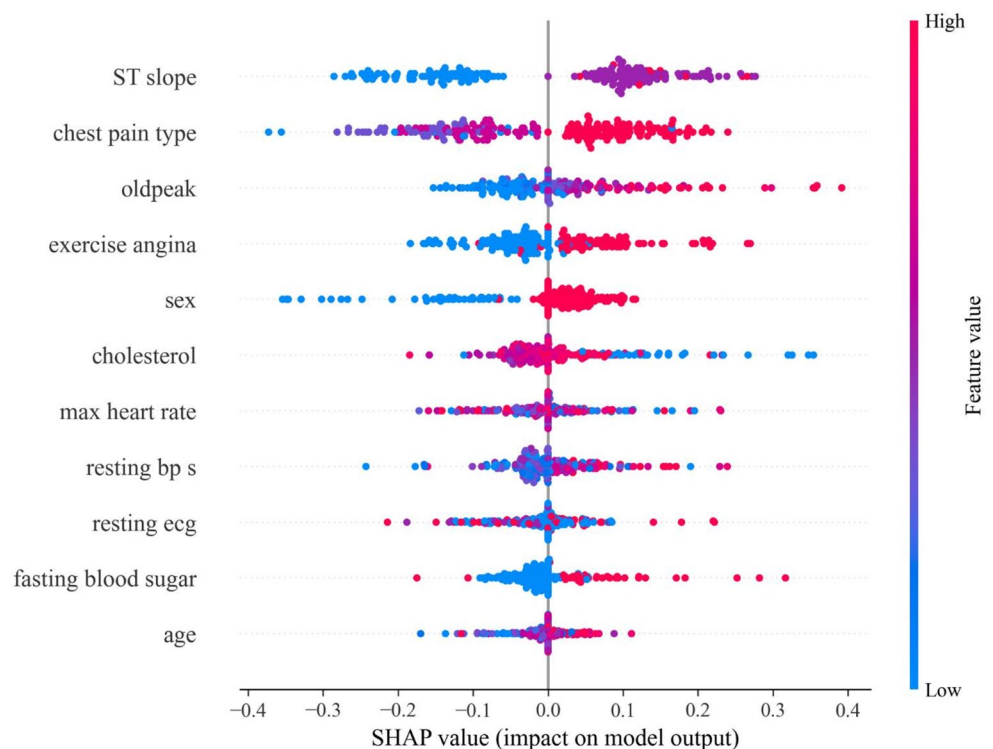


Fig. 15 SHAP summary plot - Detailed visualization of the impact of each feature on individual predictions



the electrocardiogram are a critical indicator for diagnosing heart disease. Chest pain type ranks second with a value of 0.114, followed by oldpeak (0.068), exercise angina (0.064), and gender (0.062).

Notably, cardiovascular risk factors traditionally considered important, such as cholesterol level (0.054) and

maximum heart rate (0.041), show moderate significance in the SHAP analysis. The lowest significance values are observed for age (0.024) and fasting blood glucose (0.036). These results demonstrate that the model offers a different perspective from traditional risk assessments by learning complex feature interactions.

Table 10 Performance comparison of the proposed GWO-WOA-AOA optimized CNN for different dataset configurations

Dataset	Accuracy (%)	F1_Score (%)	Recall (%)	Precision (%)	NPV (%)	MCC (%)	Kappa (%)
Original	96.22	96.41	96.03	96.80	95.58	92.42	92.41
Balanced	96.47	96.55	96.92	96.18	96.77	92.94	92.94
Double-balanced	97.42	97.42	97.22	97.61	97.22	94.83	94.83
Framingham	90.03	71.81	75.61	68.38	94.97	65.90	65.80

Fig. 16 Performance comparison of the proposed GWO-WOA-AOA optimized CNN for different dataset configurations

The SHAP summary plot presented in Fig. 15 visualizes in detail the impact of each feature on individual predictions. This plot reveals that high ST-slope values generally increase the risk of heart disease, while typical angina-type chest pain contributes similarly positively. Furthermore, gender (male=1) appears to have a significant impact on heart disease risk, with female patients generally falling into the lower-risk category.

These explainability analysis results demonstrate that the proposed model not only provides high accuracy but also produces clinically meaningful and interpretable predictions. By understanding the clinical parameters on which model predictions are based, healthcare professionals will be able to make more informed decisions regarding patient care.

4.5 Data augmentation and performance analysis on different datasets

The limited size of the dataset and class imbalance problems can negatively affect the performance of machine learning models. In this study, to address these issues, data augmentation is performed using the Synthetic Minority Oversampling Technique (SMOTE) method and the performance of the proposed GWO-WOA-AOA optimized CNN method on different dataset configurations is evaluated.

Original Dataset: The main heart disease dataset consists of 1190 patient records and contains 561 samples (47.14%)

for class 0 (no heart disease) and 629 samples (52.86%) for class 1 (heart disease). While this distribution is relatively balanced, there is a slight imbalance.

Balanced dataset: To eliminate the class imbalance in the original dataset, the SMOTE algorithm was applied, and the minority class (class 0) was brought up to the majority class level (629 samples). As a result, a balanced dataset containing a total of 1258 samples was obtained.

Double-balanced dataset: In order to increase the size of the balanced data set, SMOTE was applied again and the number of samples for both classes was doubled, creating a double-balanced data set containing a total of 2516 samples.

Framingham dataset: To test the generalizability of the method, a different source, the Framingham Heart Study dataset, was used. This dataset consists of 3658 patient records and includes 3101 negative (84.77%) and 557 positive (15.23%) samples for estimating 10-year coronary heart disease risk.

Table 10 presents the performance results of the proposed GWO-WOA-AOA optimized CNN method for four different dataset configurations. Figure 16 presents the comparative graphical analysis of the performance metrics of the proposed GWO-WOA-AOA optimized CNN method for the four different dataset configurations presented in Table 10.

An examination of the results presented in Table 10 reveals that data augmentation and balancing processes have significantly positive effects on model performance.

The results obtained with the balanced dataset (96.47% accuracy) show a slight improvement compared to the original dataset. The double-balanced dataset achieved the highest performance, demonstrating superior results across all metrics with an accuracy rate of 97.42% and an MCC value of 94.83%.

The low performance on the Framingham dataset (90.03% accuracy, 71.81% F1-score) can be explained by the dataset's different feature structure and significant class imbalance (84.77% negative, 15.23% positive). In particular, the low F1-score and MCC values indicate that the model struggles to identify the minority class (10-year CHD risk) in this dataset. However, the high NPV value (94.97%) demonstrates that the model's negative predictions are reliable.

These findings demonstrate that SMOTE-based data augmentation techniques effectively address class imbalance problems and improve model performance. In particular, the 1.2% accuracy increase and 2.41% MCC improvement achieved on the bi-balanced dataset demonstrate the practical value of data augmentation techniques. The results demonstrate that the proposed hybrid optimization approach becomes even more effective when used in conjunction with data augmentation techniques.

5 Conclusion

In this paper, a novel hybrid meta-heuristic approach based on GWO, WOA, and AOA was developed for optimizing deep CNN hyperparameters on a comprehensive and heterogeneous dataset combined from different sources, with the aim of predicting the risk of heart disease early and reliably. Unlike studies in the literature that are typically limited to a single, homogeneous dataset, this study was conducted on a large patient profile generated by combining five different datasets, highlighting the model's generalizability and clinical applicability.

The proposed hybrid GWO-WOA-AOA algorithm provides a balanced approach to hyperparameter optimization, combining exploration, exploitation, and fine-tuning. Experimental analyses on the CEC 2019 benchmark functions demonstrate that the proposed hybrid optimization algorithm produces superior and more stable results compared to both its components (GWO, WOA, and AOA) and other commonly used optimization techniques (RIME, ALO, PSO, and DE).

Optimizing the CNN model's hyperparameters with the proposed hybrid algorithm resulted in significant improvements in model accuracy (96.22%), F1 score (96.41%), sensitivity (96.03%), and precision (96.80%). Statistical significance analyses using the Wilcoxon signed-rank

test confirmed that the proposed GWO-WOA-AOA-CNN method provided statistically significant improvements compared to the reference NO-CNN method ($p=0.016$). This success demonstrated the model's robust generalization not only to single datasets but also to multi-source and highly diverse datasets, which are closer to real-world datasets. Furthermore, the low false-negative and false-positive rates demonstrate that the model minimizes the risk of misdiagnosis in clinical applications.

An explainability study conducted using SHAP analysis demonstrated that the hybrid optimized CNN model can be reliably used in clinical decision-making processes and that healthcare professionals can understand and interpret the model predictions. According to the SHAP analysis, clinical parameters such as ST slope, chest pain type, and old peak were identified as the most effective features for predicting heart disease. These findings demonstrated that the model not only achieved high accuracy but also produced clinically meaningful and interpretable predictions.

Analyses using SMOTE-based data augmentation techniques demonstrate that class imbalance problems can be effectively addressed and model performance improved. Specifically, the highest performance was achieved on the bi-balanced dataset, with 97.42% accuracy and 94.83% MCC. These findings demonstrate that data augmentation techniques become even more effective when used in conjunction with the proposed hybrid optimization approach.

Tests on the Framingham dataset, which has multiple sources, revealed that the model performs well on datasets with different feature structures, but that performance degrades in cases of significant class imbalance. 90.03% accuracy and 71.81% F1-score were achieved on the Framingham dataset, and the high NPV value (94.97%) demonstrated that the model's negative predictions were reliable.

The main contributions of this article can be summarized as follows:

- An innovative hybrid hyperparameter optimization method has been developed that combines the strengths of the GWO, WOA, and AOA algorithms.
- The CNN architecture was dynamically generated and optimized for the dataset, ensuring the model achieved the highest performance.
- The developed approach was comprehensively compared with existing literature methods, and its usability in clinical decision support systems was highlighted.
- An explainability study using SHAP analysis demonstrated that the hybrid optimized CNN model can be reliably used in clinical decision-making processes and that healthcare professionals can understand and interpret the model predictions.

- Class imbalance problems were effectively addressed with SMOTE-based data augmentation techniques, and model performance was significantly improved.
- Statistical significance analyses using the Wilcoxon signed-rank test confirmed that the performance improvements of the proposed method were not random but represented improvements in true predictive ability.
- The generalizability of the model was evaluated through tests on a multi-source Framingham dataset.

The study's limitations include: The dataset used in the study is still limited in size and feature diversity, and the model has not been tested in the real world on larger, more diverse patient groups. Furthermore, the study is the lack of a feasibility study to assess the practical applicability of the proposed system in real clinical settings. While the model demonstrates high accuracy, it needs to be evaluated in real hospital settings for performance, cost-effectiveness, system integration, healthcare professional acceptance, and regulatory compliance before being implemented in clinical settings.

Future studies plan to test the proposed method in a wider and more diverse clinical setting, further integrate explainable AI techniques, and investigate its applicability to different data types (e.g., sensor data, medical images). Furthermore, integrating the developed model into clinical processes and evaluating its practical utility with healthcare professionals will enable the method to be used as a widespread and reliable decision-support tool in healthcare. In particular, integration into real-time clinical decision-support systems, multicenter clinical validation studies, and investigation of its applicability to different types of cardiovascular diseases are among the primary objectives.

Author contributions T. L led the conceptualization, methodology development, and manuscript writing for this study. G.Y. contributed to data analysis, experimental implementation, and interpretation of the results. R.F.Ç assisted with the literature review, validation of findings, and revision of the final manuscript. All authors contributed to the work, reviewed the manuscript, and approved the final version.

Funding This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability The datasets generated and analyzed during the current study are available in the Kaggle repository.

Declarations

Competing interests The authors declare no competing interests.

Ethics declaration Not applicable.

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